



14 MW Waste-to-Energy Project by Antony Lara Renewable Energy Pvt. Ltd



Document Prepared by (LGAI Technological Center S.A. (Applus+ Certification))

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Client	Kosher Climate India Private Limited

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Summary:

The current project activity is 14MW Waste to Energy power plant at Moshi Kachara Depot, Pimpri, Pune, India. The plant utilizes Refuse-Derived Fuel (RDF) as feedstock which is derived through separation and sorting of received Municipal Solid Waste (MSW) from the Pimpri Chinchwad Municipal Corporation. The waste excludes the recyclable or potentially hazardous materials from the waste stream. Project Proponent (PD) i.e. Antony Lara Renewable Energy Pvt. Ltd (ALREPL) has an agreement with Pimpri Chinchwad Municipal Corporation (PCMC).

The main purpose of this project activity is to reduce GHG emissions through avoidance of methane production from decay of biomass through controlled combustion of MSW and renewable energy generation through steam turbine/generator. Generated electricity is supplied to national grid of India through DISCOM.

The project activity generates electricity through waste to energy-based power generation using RDF. PCMC provides around 1,200 TPD of MSW which is pre-processed in Material Recovery Facility (MRF). **After pre-processed/segregation of the MSW, 700 TPD of RDF (pre-processed) such as paper, wood including coconut shell, garden waste, cloth etc.** with applicable combustible properties has been processed by the PD - Antony Lara Renewable Energy Pvt. Ltd (ALREPL) for waste to energy plant which comes under the scope of the project activity. **The sampling of the RDF/ pre-processed waste has been used for the assessment of physical characterization of waste.** PD supplies generated electricity to Indian national grid through concession agreement with the local distribution company (DISCOM).

The project has started operation on 07-October-2023 and the project through its 7-year renewable crediting period starting from 07-October-2023 to 06-October-2030 is expected to reduce 461,756 tCO₂e emission reductions with average emission reductions as 57,719 tCO₂e¹.

The objective of this validation activity is to have an independent third-party assessment of the project design, estimated emission reductions (ER) sheet and to ensure a thorough assessment of the proposed project activity against the applicable VCS requirements, applicable CDM methodology and tools and other compliances such as;

¹ VVB noted that this value 57,719 tCO₂e is on the basis of vintage wise calculation. If annual average over the 07 year crediting period is considered then, the value would be 65,965 tCO₂e

- Baseline and monitoring methodology AMS III.E, version 17.0 and AMS I.D, version 18.0 and applicable CDM tools
- Host Country legislation and sustainability criteria along with VCS program guide version 4.4 and VCS standard version 4.7
- VCS Validation Verification Manual, version 3.2

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified carbon units (VCUs).

A risk-based approach has been followed to perform this validation activity. In the course of validation, 03 Corrective Action requests (CARs) and 05 Clarification Requests (CLs) were raised and successfully closed. 00 Forward action request (FARs) is raised for the verifying VVB.

The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, on-site visit and follow-up interviews have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to verify the fulfillment of the stated criteria of VCS.

As a part of validation, VVB has checked and confirmed project details against the submitted documents and have conducted on-site visit too and no uncertainties are associated with the validation.

Our Validation approach was based on the requirements as defined by VCS board. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these. The validation confirms that:

- The projects description compliance with relevant rules, including the Host Country legislation and sustainability criteria along with VCS program guide version 4.4 and standard version 4.7.
- The project's baseline and additionality are assessed against "AMS III.E, version 17.0 - Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment" and "AMS I.D, version 18.0 - Grid connected renewable electricity generation".
- The project's monitoring plan is assessed against "AMS III.E, version 17.0 - Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment" and "AMS I.D, version 18.0 - Grid connected renewable electricity generation".

The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, on-site visit and follow-up interviews have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to conclude positive validation opinion.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center S.A. (Hereinafter referred as Applus+ Certification) has been appointed by “Kosher Climate India Private Limited” to perform the validation of the project entitled “14 MW Waste-to-Energy Project by Antony Lara Renewable Energy Pvt. Ltd” under VCS standard, version 4.7/6/ and program guide, version 4.4/6/.

The objective of this validation activity is to have an independent third party for the assessment of the project design and other documents to ensure a thorough assessment of the proposed project activity against the applicable VCS requirements. In particular;

- The project’s baseline is assessed against CDM approved methodology – AMS III.E, version 17.0/4/ and AMS I.D, version 18.0/4/.
- The project’s monitoring plan is assessed against CDM approved methodology – AMS III.E, version 17.0 and AMS I.D, version 18.0.
- The projects compliance with the requirements of relevant rules and requirements, including the Host Country legislation and sustainability criteria along with VCS requirements/6/ and CDM approved methodologies/4/ and applied tools/5/.
- VCS Validation and Verification manual version 03.2/6/

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified carbon units (VCUs).

1.2 Scope and Criteria

The scope of validation is independent and objective review of the submitted VCS Project Description (VCS PD)/1/ against the relevant criteria and decisions (see 1.1).

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the VCS PD. The main focus of the assessment team is to identify the significant risks for the project description, implementation and the generation of VERs. The validation activity is not meant to provide any consulting to the project proponents however, stated requests for clarifications and corrective actions may have provided input for improvement of the project design.

The only purpose of the validation is its usage during the registration process as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can’t be held liable by any party for decisions made or not made based on the validation opinion, which will go beyond that purpose.

1.3 Reasonableness of Assumptions

The validation has been planned and organized to achieve a Reasonable Level of assurance as per the requirement of VCS.

1.4 Summary Description of the Project

The project involves the development of a 14MW Waste-to-Energy Power Plant located at Moshi Kachara Depot, Pimpri, Pune, India. This facility utilizes Refuse-Derived Fuel (RDF) obtained from the separation and sorting of Municipal Solid Waste (MSW), excluding recyclable and potentially hazardous materials. The MSW, sourced from the Pimpri Chinchwad Municipality area under a concession agreement with the Pimpri Chinchwad Municipal Corporation (PCMC), is pre-processed at a Material Recovery Facility (MRF). The PCMC supplies approximately 1,200 tons per day (TPD) of MSW, from which 700 TPD of RDF, with suitable combustible properties, is processed by Antony Lara Renewable Energy Pvt. Ltd. (ALREPL)/11/ for the Waste-to-Energy plant and inert waste is disposed of in a sanitary landfill located within the same area.

ALREPL is responsible for the construction and operation of this RDF-based Waste-to-Energy plant/11/. The primary objective of this project is to generate clean electricity by converting waste into energy. The plant will process around 700 TPD of RDF which includes (pre-processed) waste such as paper, wood including coconut shell, garden waste, cloth etc., generating 14MW of electricity/14/. It employs a reciprocating grate firing type boiler with a 70 TPH steam generation capacity and a 14MW double bleed cum condensing steam turbine generator/16//17/, which is connected to the Indian grid to supply 77,616 MWh/year of net electricity/2/. PCMC has signed an agreement/17/ to utilize the electricity generated by the project through a grid connectivity.

The project aims to mitigate methane emissions from landfills by converting Municipal Solid Waste (MSW) into renewable electricity, which replaces electricity that would otherwise be sourced from grid-connected power plants. Over the 7-year crediting period, the project is expected to achieve an average annual emission reduction of 57,719 tCO₂e², totaling 461,756 tCO₂e/2/. Prior to the project activity, MSW from the PCMC area was primarily disposed of in a sanitary landfill without methane recovery, leading to methane emissions from anaerobic decay. And, the electricity would have been predominantly supplied by grid-connected power plants dominated by fossil fuel based units³. This greenfield project will reduce methane emissions through treatment of waste for electricity generation thus, reducing greenhouse gas emissions.

² VVB noted that this value 57,719 tCO₂e is on the basis of vintage wise calculation. If annual average over the 07 year crediting period is considered then, the value would be 65,965 tCO₂e

³ <https://powermin.gov.in/en/content/power-sector-glance-all-india>

2 VALIDATION PROCESS

2.1 Method and Criteria

Validation scope: The scope is defined as an independent and objective review of project design document. The VCS PD was reviewed against the criteria stated in VCS guidance documents, approved CDM baseline and monitoring methodologies AMS III.E, version 17.0 and AMS I.D, version 18.0. The validation was based on the requirements in the VCS validation and verification manual, version 3.2, VCS standard, version 4.7, VCS program guide, version 4.4.

The validation is not meant to provide any consulting towards the project proponents. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project document.

Validation process: The project assessment is based on the VCS validation and verification manual, version 3.2, VCS standard, version 4.7, VCS program guide, version 4.4 and is conducted using standard auditing techniques to assess the correctness of the information provided by the project proponents.

The Validation process has been conducted following the regulatory documents and based on the ISO 14064-3:2019 and ISO 14065:2020 using standard auditing techniques as required by the VCS Standard version 4.7.

The validation process has been conducted through an onsite inspection (see below), thus, no need for a risk analysis for remote inspections has been determined for this validation (cf. VCS Standard v4.7 Para 4.1.13).

No ICT means have been used for the conduction of this validation process other than those for the provision of documents and transferring information and undertake normal communications with the PP(s). Those ICT means supporting the process in this sense, have been used to an extent sufficient to cover the background necessities of validation process and have demonstrated enough effectiveness to conduct it in accordance with the relevant applicable regulatory requirements.

Before the assessment begins, members of the team covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed in accordance with the Applus+ Certification's procedures and policies, that are in accordance with the ISO 14065:2020 regulatory provisions.

Applus+ Certification has completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of:

- Protocols used to estimate or measure GHG emissions from these sources;
- Collection and handling of data;

- Controls on the collection and handling of data;
- Means of validating the calculated data; and
- Compilation of the information in the VCS PD.

Based on the above, the VVB prepared a validation plan and applied an evidence-gathering plan designed to collect sufficient and appropriate evidence upon which to base the conclusion and to consider the inherent risk for the validation. The evidence-gathering plan is based on analytical procedures and tests and considers the thresholds for materiality as per the regulatory documents applied for this validation (5% as per the applied VCS Standard version 4.7).

The applied evidence-gathering plan is used by the VVB to determine the conformity of the statements made by the client against the principles and requirements of the applied regulatory documents during this validation.

The evidence-gathering plan designed by the VVB takes into consideration, as a minimum:

- The selection and management of the data and information;
- The processes for collecting, processing, consolidating, aggregating and reporting the data and information;
- Systems and processes in place to ensure the validity and accuracy of the data and information;
- The design and maintenance of the system to control the data and information;
- Systems, processes and personnel that support system in place, including activities for ensuring data quality;
- The plans for instrument maintenance and calibration where appropriate;

Once the project's initial documentation is received by the assessment team, the members of the assessment team carried out:

- A desk review of the VCS PD/1/;
- On-site Audit conducted with project proponent and stakeholders;
- The resolution of outstanding issues and the issuance of the final validation report (FVR) and validation opinion.

The prepared validation report and other supporting documents then undergo an internal quality control at the HQ (Accredited office) before being submitted to the VCS executive board.

In order to ensure transparency, assumptions must be clear and stated explicitly and background material must also be referenced.

Appointment of the assessment team:

According to the sectoral scope / technical area and experience in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of audit team shall be approved by the LGAI Technological Center S.A. (Applus+ Certification) ensuring that the required skills are covered by the team.

The qualification levels for team members that are assigned by formal appointment rules are as presented below:

- Lead Auditor (LA)
- Auditor (A) / Auditor in Training (AiT)
- Financial Expert (FE)
- Technical Expert (TE)
- Technical Expert in Trainee (TEiT)
- Technical Reviewer (TR)
- Other Supporting roles such as Financial Expert (FE) or Country Expert (CE) /Local Expert (LE)
- Any of the above-mentioned roles in training (iT, e.g. AiT for auditor in training, TEiT for Technical Expert in Training, etc.).

The sectoral scope / technical area knowledge linked to the applied methodology/ies shall be covered by the assessment team.

Name	Role	SS Coverage	TA Coverage	Financial aspect	Host country experience
Mr. Deepak Pundlik	LA / TE (1.1) / TEiT (13.1)	Yes	Yes (for both)	No	Yes
Dr. Atul Takarkhede	FE / TE (13.1)	Yes	Yes (for both)	Yes	Yes
Mr. Nikunj Agarwal	TR	Yes	Yes (for both)	Yes	NA

The complete list of CVs is included as Appendix 4 of this report.

Document Review: The VCS PD/1/ submitted by the PP was reviewed against the approved methodology/4/ and other relevant criteria/5//6/ to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from other sources has been done. A complete list of all documents and evidence material reviewed is included in this report below in appendix 1.

Follow -up Interviews: A site visit is conducted by LGAI Technological Center S.A. (Applus+ Certification) who performed interviews and physical site inspection with project stakeholders to confirm the presented information and to resolve issues identified in the document review. The detail is provided in this report in the below sections.

Resolution of Clarification and Correction Action Request: The objective of this validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification) positive conclusion on the project design document. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the PP and Applus+ Certifications to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the appendix 3.

Validation schedule:

Sr. No.	Description	Date
1	Desk review of the documents	02-March-2024
2	Onsite visit	15-March-2024 to 16-March-2024
3	Release of Findings	01-April-2024

Resolution of Clarification and Corrective Action Request

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for Applus+ Certification positive conclusion on the submitted VCS PD. The Corrective Action Requests and Clarification Requests raised by Applus+ Certification were resolved during communications between the Client and Applus+ Certification to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the Appendix 3.

The final PD submitted by PP, serves as the basis for the final assessment presented. Additional changes to the project during the verification process are not considered to be significant with respect to the main VCS objectives. The two VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

The most recent version of VCS PD/1/ submitted by PP serves as the basis for the final assessment presented.

Internal quality Control: As final step of a validation, the final documentation have to undergo an internal quality control by the technical review committee, i.e., each report has to be finally approved either by the head of the technical review committee or the deputy. In case one of these two persons is part of the assessment team approval can only be given by the other one to avoid any conflict of Interest.

After confirmation of the project proponents, the positive validation/verification opinion and relevant documents are submitted to the VCS secretariat through the VCS web-platform.

2.2 Document Review

The details of the document observed during validation process are listed below in appendix 1 of this report.

2.3 Interviews

An on-site visit was conducted for the project activity on 15/03/2024 and 16/03/2024.

VVB validated technical details/20/ & monitoring arrangements along with project's operation including technical details, monitoring records, and calibration certificates shared by PP. All the documents were cross checked to ensure conservative estimation of emission reductions has taken place.

During on-site audit, the PP representatives were questioned about the implementation of the project activity. Several topics like the waste collection, separation and then feeding to boiler, boiler operations, turbine specifications and operation, energy meters, waste weighing, waste characterization, data recording and monitoring, error accountability were discussed. To cross check the information provided by PP, various documents like technical specifications, commissioning certificates, statutory clearances, PPA, JMR records, steam generation data, waste collection and disposal data, invoices, equipment calibration records etc. were also verified. The names of the persons interviewed during the audit is given below;

Sr. No.	Name	Designation and company	Interviewed by
1	Mr. Ashutosh Rana	Sr. Manager, ALREPL	Mr. Deepak Pundlik and Dr. Atul Takarkhede
2	Mr. Santosh Pawar	Representative, PCMC	
3	Mr. Amol Yadav	Sr. Engineer, ALREPL	
4	Ms. Surekha Kudachi	Ragpicker, PCMC waste dumping gorund	
5	Ms. Asha bai	Ragpicker, PCMC waste dumping gorund	
6	Mr. Suresh Joshi	Security assistant, PCMC Moshi landfill	
7	Ms. Harini	Sr. consultant, Kosher Climate	

2.4 Site Visits

Duration of on-site visit: 15/03/2024 and 16/03/2024				
No.	Activity performed on-site	Site location	Date	Team member
1.	Assessment team checked the implementation of the project, Baseline emissions, Emission reduction calculation, technical description of the project and Monitoring, stakeholder grievances.	Moshi, Pune, India	15/03/2024 & 16/03/2024	Mr. Deepak Pundlik and Dr. Atul Takarkhede

2.5 Resolution of Findings

The objective of this validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on the project design document. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the PP and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the Appendix 4.

Following is the list of findings which were raised and successfully closed.

Areas of validation findings	No. of CL	No. of CAR	No. of FAR
Project Details	CL01, CL05	CAR 01	-
Safeguards	-	-	-
- No Net Harm	-	-	-
- Local Stakeholder Consultation	CL02, CL03	-	-
- Environmental Impact	-	-	-
- Public Comments	-	-	-
- AFOLU- Specific Safeguards	-	-	-
Application of Methodology	-	-	-
- Title and Reference	-	-	-
- Applicability	CL04	-	-
- Project Boundary	-	-	-
- Baseline Scenario	-	-	-
- Additionality	-	CAR 02	-
- Quantification of GHG Emission Reductions and Removals	-	CAR 03	-
- Methodology Deviations	-	-	-
- Monitoring Plan	-	-	-
Non-Performance Risk Analysis	-	-	-
Others (please specify)	-	-	-
Total	05	03	00

2.5.1 Forward Action Requests

No FAR is raised during this validation.

3 VALIDATION FINDINGS

3.1 Project Details

The current project activity is 14 MW Waste to Energy power plant in Moshi, Pune, Maharashtra state, India/11//12/15/. The plant utilizes Refuse-Derived Fuel (RDF) as feedstock which is derived through separation and sorting of received Municipal Solid Waste (MSW) from the Pimpri Chinchwad city. The waste excludes the recyclable or potentially hazardous materials from the waste stream. Project Proponent (PD) i. e. Antony Lara Renewable Energy Pvt. Ltd. (ALREPL) has an agreement/17/ with Pimpri Chinchwad Municipal Corporation for the disposal of collected waste after processing it for RDF and generating electricity from this RDF.

The main purpose of this project activity is to reduce GHG emissions through avoidance of methane production from decay of biomass through controlled combustion of MSW and renewable energy generation using steam turbine/generator which is supplied to national grid of India through DISCOM/21/.

The project activity involves R grate incineration technology to incinerate RDF prepared from MSW sourced from Pimpri-Chinchwad city that would otherwise be disposed of at a local landfill site. The main equipment in the project activity includes 1 incinerator (700 ton of RDF per day), 1 boiler (70 tons per hour), 1 steam turbine and 1 generator (14 MW), the transformer station and electricity transmission line, exhaust gas cleaning systems and ash removal systems.

The steam generating system of the Project activity consists of a RDF fired incinerator boiler with a Maximum Continuous Rating (MCR) of 1 x 70 TPH, with the outlet steam parameters at 45 kg/cm²g and 410 °C. The control range on the super-heater outlet temperature shall be $\pm 50^{\circ}\text{C}$. The combustion system of the boiler is stoker type. The boiler efficiency when firing 100% waste will be a minimum of 68 % on the Gross Calorific Value (GCV) basis. The dust concentration in the flue gases leaving the boiler will be a maximum of 50 mg/Nm³. The grate acts as a continuous ash discharge and an overfeed stoker, with hydraulic drive. Fuel is burnt, suspended in air above the grate and on the traveling grate. Ash generated in the boiler is continuously discharged. Air seals are provided in order to control the excess air entry in to the furnace.

The project has started operation on 07-October-2023 as confirmed based on commissioning of boiler/turbo generator/14/ and permission to connect to grid/10/. The project through its 7-year renewable crediting period starting from 07-October-2023 to 06-October-2030 is expected to reduce 461,756 tCO_{2e} emission reductions with average emission reductions as 57,719 tCO_{2e}.

This is a greenfield project activity as confirmed during the on-site interviews and based on concession agreement/17/ between Pimpri Chinchwad Municipal Corporation and PP. Prior to the implementation of the project activity, MSW in Pimpri Chinchwad city was taken to the designated site to landfill in anaerobic condition without any methane recovery system for flaring or energy recovery purpose which was confirmed during site visit to the landfill area which is located next to the project location. Therefore, in the absence of the project activity, methane emissions from anaerobic decay of waste would have been

released into the atmosphere. For electricity component, prior to the implementation of the project activity, the equivalent amount of electricity would have generated through grid connected power plants which are dominated by fossil fuel-based units⁴.

This is a standalone project implemented by ALREPL and is not a grouped project.

The project proponent for the project is Antony Lara Renewable Energy Pvt. Ltd. (ALREPL)/11//12/ and other entity involved in the project is Kosher Climate India Private Limited. The details of both the entities are as below:

Organization Name	Antony Lara Renewable Energy Pvt. Ltd. (ALREPL)
Contact person	Mr.Jose Jacob Kallarakal
Address	Row House No. 6, Kanakia Spaces, Teen Hath Naka, Near TATA Motors Thane(W) 400604
Telephone	0224-2130300
Email	compliance@antonywaste.in

Organization Name	Kosher Climate India Private Limited
Contact person	Mr. Narendra Kumar Ramaraj
Address	11-15/13, Srinivas Nagar Colony, Nagaram, ECIL, Hyderabad, Telangana - 500083, India.
Telephone	+91-9632803444
Email	narendra@kosherclimate.com

The project ownership is with Antony Lara Renewable Energy Pvt. Ltd. (ALREPL) as confirmed based on commissioning certificate/14/, statutory clearances by local pollution control board/14/ and agreement with PCMC/17/.

The project start date is 07-October-2023 as confirmed based on commissioning certificate/14/ copy submitted. VVB through on-site visit and interviews has confirmed that on this day MSW was collected, processed to produce RDF and started generating electricity through combustion.

The lifetime of the project is considered 21 years/20/. The boiler and turbo-generator being the main equipment which is used to burn RDF leading for electricity generation. This is found acceptable based on manufacturer's specifications submitted.

Chronology of events of the project activity is provided below which was checked and confirmed based on the evidence submitted i. e. DPR, incorporation certificate, copy of concession agreement etc.

⁴ <https://powermin.gov.in/en/content/power-sector-glance-all-india>

S. No	Activity	Date of Activity
1.	Request for proposal by PCMC for setting up waste to energy plant at existing landfill site at Moshi	25-September-2017
2.	Preparation of Techno-commercial document by ALREPL	10-January-2018
3.	Bid submission by ALREPL (Investment decision date)	11-January-2018
4.	Letter of award to ALREPL	28-February-2018
5.	Signing of Concession Agreement by ALREPL with PCMC	06-September-2018
6.	Signing of Conversion charges Agreement	26-June-2020
7.	Signing of EPC Contract by ALREPL	06-July-2020
8.	Commissioning of boiler to fire RDF processed	07-October-2023

The project has opted for renewable type of crediting period of 7 years. Since this is a Non AFOLU project, it has an option to choose between the above and fixed 10-year crediting period. Hence, the chosen one as renewable period is found acceptable.

The project start date is 07-October-2023 since as observed during the on-site visit, the project has started generating electricity on this date and sending the same to the electricity grid thus it started emission reductions. Thus, the 1st renewable 7-year crediting period will be from 07-October-2023 to 06-October-2030.

The expected emission reductions over 1st crediting period are as below:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
07-October-2023 to 31-December-2023	4,374
01-January-2024 to 31-December-2024	22,685
01-January-2025 to 31-December-2025	39,681
01-January-2026 to 31-December-2026	55,788
01-January-2027 to 31-December-2027	70,992
01-January-2028 to 31-December-2028	85,550
01-January-2029 to 31-December-2029	99,191
01-January-2030 to 06-October-2030	83,495
Total estimated ERs during the first crediting period	461,756
Total number of years	07

Average annual ERRs

57,719

The Project is located at Tapkir Nagar, Moshi Pimpri Chinchwad 412105, Maharashtra, India. The geo-coordinate of the project are 18°39'29.52"N, 73°51'26.51"E, which were cross checked with the help of google maps and confirmed during the on-site visit.

Project description and conditions prior to project initiation:

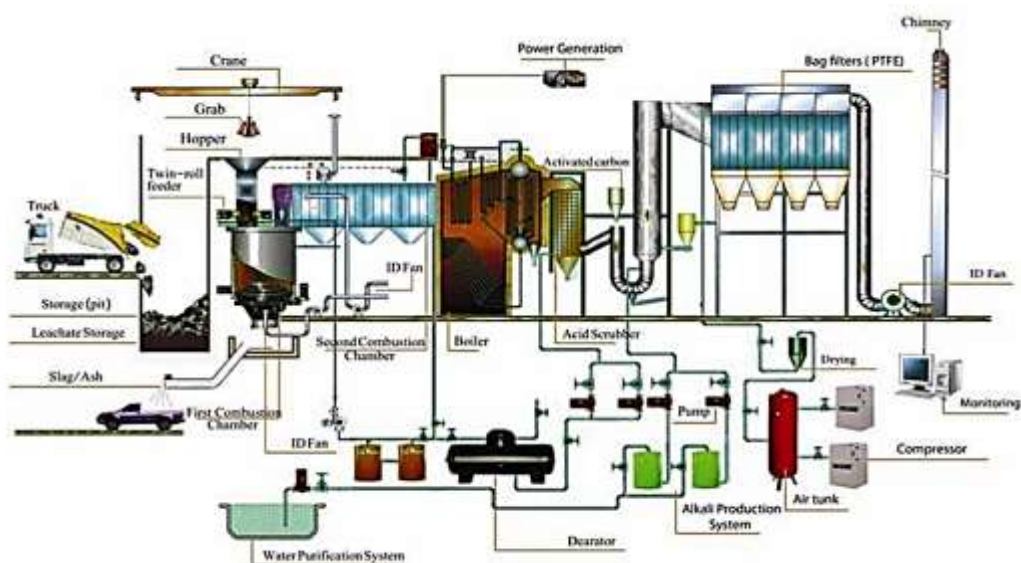
VVB has confirmed the project design during on-site visit and based on submitted documents such as, consent to establish / operate by local pollution control board/11/, technical specifications sheets for the equipments/20/, concession agreement/17/, single line diagram/8/, MSW records maintained at site/32/, energy generation records/21/, operation and maintenance manual/24/ etc.

The project activity involves R grate incineration technology to incinerate RDF prepared from MSW which is collected from Pimpri Chinchwad city. In baseline scenario, the collected MSW was getting disposed of at a local landfill site.

The main equipment in the project activity includes 1 incinerator (700 ton of RDF per day), 1 boiler (70 tons per hour), 1 steam turbine and 1 generator (14 MW), the transformer station and electricity transmission line, exhaust gas cleaning systems and ash removal systems.

Project operations: VVB during on-site visit confirmed the project operation and a brief description is presented here. The fresh MSW received at the unloading bay is fed to the pre-processing section. The waste is then subjected to the moving bed feeder for further process, which includes splitter, start screen, shredder. The treated waste post this section is then called as Refused Derived Fuel (RDF). After pre-processing, the RDF is sent to the waste feeding section. RDF is fed to the boiler by grabber through vibro feeder at the rate of 14 TPH. RDF is incinerated on travelling grate. The unburnt components continue to move on the grates to get out of the combustion chamber and fall into the slag collector of furnace bottom in form of bottom ash. The heat recovered from the incineration process will be provided to boilers. The superheated steam generated from the boilers will be fed into the steam turbine to generate kinetic energy. Turbines are connected to generators to produce electricity. The electricity generated from the project activity will be exported to the national grid through the electricity transmission line. The flue gases from the incineration process including NO_x, SO_x, HCl, and dioxin or furan if any are treated by the flue gas cleaning technology. This includes combustion control, electrostatic precipitator, lime power dosing, activated carbon adsorption, bag filter. After the treated clean gas, which meets states and national Environmental Quality Standard is released to the atmosphere from chimneys. The bottom ash is collected and sent to the brick manufactures while fly ash will be used for paver production. However, as on date it is sent to brick manufacturers.

The layout of the plant is shown in below figure:



The project activity does not fall under the purview of the Environmental Impact Assessment (EIA) notification of the Ministry of Environment, Forest & Climate Change (MOEF&CC), Government of India (As per MOEF&CC Notification No. - S.O. 1533, dated 14-September-2006⁵). However, PP has conducted Environment and Social Impact Assessment in October, 2021/7/ which was checked and confirmed. The project has obtained Consent to Establish (CTE)/11/ and Consent to Operate (CTO)/11/ from the state Pollution Control Board under Air (Prevention and Control of Pollution Act, 1981⁶ and Water (Prevention and Control of Pollution) Act 1974⁷ which were checked and confirmed. The project activity is implemented in compliance to The Factories Act, 1948⁸ & The Indian Boilers Act-1923⁹ as confirmed onsite based on copies of clearances submitted. The project is in compliance with the Municipal Solid Waste (Management and Handling) Rules 2000¹⁰, where all the municipal authorities in country are directed to manage solid waste in their respective jurisdiction and is implemented in compliance.

Thus, VVB confirmed that the project activity is in compliance with all relevant statutory and regulatory laws applicable in the host country.

VVB has checked other carbon registries such as CDM, GS4GG, GCC and I-REC to confirm that the project activity is neither registered nor applied for registration with any of these GHG or Non GHG program(s) to obtain the certified GHG emission reductions. PP has submitted a declaration also that the emission reductions claimed under this project activity will not be double counted under any other GHG program.

⁵ https://environmentclearance.nic.in/writereaddata/EIA_Notifications/1_SO1533E_14092006.pdf

⁶ <https://cpcb.nic.in/displaypdf.php?id=aG9tZS9haXltcG9sbHV0aW9uL0dTU02RS5wZGY=>

⁷ https://maitri.mahaonline.gov.in/pdf/The_Water_Prevention_and_Control_of_Pollution_Act_1974.pdf

⁸ https://environmentclearance.nic.in/report/EIA_Notifications.aspx

⁹ <https://www.ilo.org/dyn/natlex/docs/ELECTRONIC/48110/114241/F-1729379182/IND48110.pdf>

¹⁰ https://cpcb.nic.in/uploads/MSW/MSW_AnnualReport_2001-02.pdf

VVB checked and confirmed that the project activity is neither registered nor rejected by any other GHG program.

The current project does not involve any other form of credits and supply chain (scope 3) emissions.

Sustainable Development Contributions: VVB noted that the plant operation directly covers key areas of sustainable development including renewable power generation, solid waste management, wastewater management and efficient usage, environment protection, innovative technological integrations, and partnership for sustained growth.

Among 17 SDGs, PP has claimed 07 SDGs which are directly linked with the project location operation and innovations in the project operations.

1. Clean water and sanitation (SDG-6)

Goal: Ensure availability and sustainable management of water and sanitation for all

Target: 6.3 By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally.

Indicator: 6.3.1 Proportion of domestic and industrial wastewater flows safely treated

ALREPL shall employ all water conservation methods to ensure that the plant does not consume any fresh water (Zero Fresh Water Intake plant).

Monitoring Parameter: Net Inlet Quantity of sewage treated water per day (100m³/day)

2. Affordable and clean energy (SDG-7)

Goal: Ensure access to affordable, reliable, sustainable and modern energy for all

Target: 7.2: By 2030 increase substantially the share of renewable energy in the global energy mix

Indicator: 7.2.1: Renewable energy share in the total final energy consumption

The plant capacity of 14MW generates renewable power. To calculate the carbon emission that would have otherwise generated from fossil fuel energy generation and also to substantiate SDG 7. Net quantity of renewable energy generated from the power plant, which otherwise would have been generated from the grid connected power plants.

Monitoring Parameter: Net Electricity Generation (GWh/year)

3. Decent work & economic growth (SDG-8)

Goal: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

Target: 8.5 By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value

Indicator: 8.5.1 Average hourly earnings of employees, by sex, age, occupation and persons with disabilities

The project activity provides employment opportunities and the average salaries has been paid to employees irrespective of sex, age group, etc which leads to increased productivity, driving economic development. By additional revenue generation from municipal waste, post combustion ash and recovering valuable metals and non- metals from the waste.

Monitoring Parameter: No of employment (with bifurcation on number by sex, age group and where applicable, persons with disabilities), Average earnings, Policy for Non-discrimination and equal pay for the work of equal value.

4. Sustainable cities and communities (SDG-11)

Goal: Make cities and human settlements inclusive, safe, resilient and sustainable

Target: 11.6 By 2030, reduce the adverse per capita environmental impact of cities, including by paying special attention to air quality and municipal and other waste management

Indicator: 11.6.1 Proportion of municipal solid waste collected and managed in controlled facilities out of total municipal waste generated, by cities

Monitoring Parameter: Net Inlet Quantity of Municipal Solid Waste (Tonnes per day)

5. Responsible consumption and production (SDG-12)

Goal: Ensure sustainable consumption and production patterns

Target: 12.5 By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse. By adopting material conservation and waste recycling approaches in plant design and operations. Recovery of metal and non-metal waste from MSW for re-using to boost circular economy.

Monitoring Parameter: Net Inlet Quantity of Municipal Solid Waste (Tonnes per day), Quantity of scrap metal recovered (Tonnes per day)

6. Climate action (SDG-13)

Goal: Take urgent action to combat climate change and its impacts

Target: 13.2 Integrate climate change measures into national policies, strategies and planning

Indicator: 13.2.2 Total greenhouse gas emissions per year. Reduction of annual average of 57,719 tons of CO₂ eq. greenhouse gas (GHG) emissions per year by coal replacement and avoiding open dumping of waste. Project activity generates renewable energy-based electricity and mitigates the CO₂ emissions which would have been generated from the grid-connected power plants. Project has already commissioned and started reducing the emissions. Hence, complied to the SDG 13

Monitoring Parameter: GHG emission reductions (tCO₂/year)

7. Partnership for the goals (SDG-17)

Goal: Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development

Target: 17.17 Encourage and promote effective public, public-private and civil society partnerships, building on the experience and resourcing strategies of partnerships. By setting up of WTE plant in PPP (Public-Private Partnership) between M/s Antony Lara Renewable Energy Pvt. Ltd (ALREPL) and PCMC (Pimpri Chinchwad Municipal Corporation).

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Audit history	In section 1.2 of the submitted PD, PP has provided audit history. The same was checked and found appropriate.
Sectoral scope	13.1 & 1.1
AFOLU project category, if applicable	NA
Project activity type	Type I and Type III Project is not a part of grouped project as confirmed during the on-site visit.
General eligibility of the project to participate in the VCS Program	VVB team verified that; • This project activity is not excluded under Table 1 in section 2.1.3 of the VCS Standard v4.7. • This project activity is non-AFOLU project and completing validation within two years of the project start date as per clause 3.8.1 of VCS Standards 4.7. • The project is not a fragmented part of a larger project or activity that would otherwise exceed such limits as it is the only project activity as confirmed during the on-site visit. • The opening meeting with the VVB took place only after the project had been placed under validation in the VCS Registry. During desk review, VVB team found that project proponent submitted project description and accomplice document to VERRA for project pipeline listing afterwards project was open for public comments for the period from 21-November-2023 to 21-December-2023¹¹. VVB team verified that, the

¹¹ <https://registry.terra.org/app/projectDetail/VCS/4620>

	project activity is Waste-to-Energy Power Plant and applied methodologies AMS III.E, version 17.0 and AMS I.D, version 18.0 and the applied methodology is eligible under the VCS Program. The applicability condition of the applied methodology is found consistent in section 3.4 of this joint validation and verification report.
AFOLU project eligibility, if applicable	Not applicable
Transfer project eligibility, if applicable	Not applicable, as the project is not being transferred from any other approved GHG Program. Declaration for same has been provided by Project Proponent mentions that the project is not registered/seeking registration in any other GHG registry apart from Verra. VVB has verified the declaration/30/ and checked various registries including GS, GCC, CDM etc/26//27//28//29/. and found that the project is not registered in any of them and does meet the criteria mentioned in Appendix 2 and Section 3.22 (Participation under other GHG Programs) of the VCS Standard v4.7.
Project design	As per para 3.6 of the VCS Standard version 4.7, the proposed project activity is single location project.
Project ownership	Antony Lara Renewable Energy Pvt. Ltd. (ALREPL) as confirmed based on commissioning certificate, letter of award issued to PP and hence acceptable.
Project start date	07-October-2023 which is confirmed based on the copy of commissioning certificate submitted.
Project crediting period	07-October-2023 to 06-October-2030 i. e. 07 years with twice renewable which is in line with VCS standard, version 4.7 and acceptable.
Project scale	The assessment team has crosschecked and found that the estimated annual emission reduction 57,719 tCO ₂ ¹² as per para 3.10.1 (1) VCS standard version 4.7, the emission reduction of project activity is less than 300,000 tCO ₂ hence project is not large-scale project and will be considered as Project.

¹² VVB noted that this value 57,719 tCO₂e is on the basis of vintage wise calculation. If annual average over the 07 year crediting period is considered then, the value would be 65,965 tCO₂e

Likelihood of achieving estimated GHG emission reduction or removals	During review, VVB term observed that estimated emission reduction calculated based on waste to energy project. PP submitted ER sheets and estimated annual emission reductions over 07 year crediting period are 57,719 tCO ₂ e/year and project activity will reduce total of 461,756 tCO ₂ e during the first crediting period from 07-October-2023 to 06-October-2030 (inclusive of both the dates). Thus, VVB confirm chance of likelihood of achieving estimated GHG emission reduction or removals.
Technologies and measures implemented by the project activity or activities	The project activity involves a 14 MW Waste to Energy (WTE) power plant located at Moshi Kachara Depot, Pimpri, Pune, India. The plant, operated by Antony Lara Renewable Energy Pvt. Ltd. (ALREPL), uses Refuse-Derived Fuel (RDF) sourced from the separation and sorting of Municipal Solid Waste (MSW) provided by the Pimpri Chinchwad Municipal Corporation (PCMC). PCMC supplies approximately 1,200 tons per day (TPD) of MSW, which is pre-processed in a Material Recovery Facility (MRF). Of this, 700 TPD of RDF, with suitable combustible properties, is used by ALREPL for energy generation.
Implementation schedule of the project activity or activities	The assessment team confirmed that the project activity was implemented on 07-October-2023, marking the commissioning of the MSW processing facility. On this date, the facility began collecting and processing municipal solid waste (MSW) to produce refuse-derived fuel (RDF) and started generating electricity through combustion.
Project location	Project is located in PCMC Garbage depot, Tapkir Nagar, Moshi Pimpri Chinchwad 412105, Maharashtra, India. The geographical co-ordinates which were checked during on-site visit using a mobile software and cross checked on google map are 18° 39' 29.52" N, 73° 51' 26.51" E.
Conditions prior to project initiation	The assessment team during on-site visit verified that, prior to the implementation of the project activity, untreated waste was disposed of in the PCMC landfill, leading to methane emissions from anaerobic degradation. This finding was corroborated during onsite interviews with local villagers and electricity would have been generated by grid connected power plants.

Project compliance with applicable laws, statutes and other regulatory frameworks	The assessment team confirms this for the host country, India. Furthermore, the team verifies that while there are no specific laws or regulations governing waste-to-energy projects in India, the project complies with all relevant existing laws and regulations.
Double counting and participation under other GHG programs	The project is seeking registration under VCS (with Project ID VCS 4620) only. Assessment team further added that, project activity is not receiving or not seeking any credit for reductions and removals from a project activity under another GHG program i.e. CDM, GS4GG, GCC. The project proponent has submitted the no double accounting declaration/30/ about non-participation under other GHG program which is checked and found appropriate. Project has provided the self-declaration/30/ that, project has not gone under registration process in any GHG/Non-GHG registry other than VCS and project activity is not rejected by other mechanism Furthermore, the project proponent has submitted documentation/30/ affirming that the net GHG emission reduction generated by the project activity will not be utilized for compliance with any other emission trading program or to meet binding limits on GHG emissions. Independently, the assessment team verified that the project is not participating in GHG programs other than the VCS program by checking other registries.
No double claiming with emissions trading programs or binding emission limits	The project activity is not participating in other emission trading programs or binding emission limits. The letter of undertaking has been furnished by the project proponent confirming that net GHG emission reduction or removal generated by the project will not be used for compliance with an emission trading programme or to meet the binding limit on GHG emission. There is no emissions trading programs or binding emission limits in State of Maharashtra, India.
No double claiming with other forms of environmental credit	VVB has noted that project GHG reductions from this project activity are not part of any environmental credit scheme and the project activity has not sought, received, or is planning to receive credit from another GHG-related environmental credit system.

	PP has provided a declaration/30/ that the project activity has not sought, received, or is planning to receive credit from another GHG-related environmental credit system.
Supply chain (Scope 3) emissions double claiming	During Desk Review and document review, VVB team confirmed that the project proponent i.e., M/s Antony Lara Renewable Energy Pvt. Ltd (ALREPL) is not buyer or seller of a product whose emissions footprint is changed by the project activities. Hence Accepted.
Sustainable development contributions	Clean water and sanitation (SDG-6): ALREPL is committed to ensuring the sustainable management of water and sanitation by implementing comprehensive water conservation strategies that align with the SDG 6.3 target of improving water quality by 2030. The plant is designed to operate as a Zero Fresh Water Intake facility, relying solely on treated sewage water with a monitored daily intake of 100m³. VVB has reviewed this approach and concludes that it effectively supports the global objectives of reducing pollution, minimizing hazardous chemical releases, and promoting safe water reuse. ALREPL's initiative is consistent with best practices in water conservation and contributes to the broader goal of sustainable water management. Affordable and clean energy (SDG-7): The project aligns with the goal of ensuring access to affordable, reliable, and sustainable energy by contributing to SDG Target 7.2, which aims to substantially increase the share of renewable energy in the global mix by 2030. With a 14 MW capacity, the plant generates renewable electricity, thereby reducing carbon emissions by replacing energy that would have been produced by fossil fuel-based grid power plants. The project's impact is measured through the net quantity of renewable energy generated annually in GWh. The Validation and Verification Body (VVB) has assessed the project and confirms that it supports the increase of renewable energy in total final energy consumption. The project's monitoring framework effectively tracks its contribution to reducing fossil fuel dependence and advancing global energy sustainability, making it a key contributor to achieving the 2030 renewable energy targets.

Decent work & economic growth (SDG-8): The project supports the goal of promoting sustained, inclusive, and sustainable economic growth by providing employment opportunities that contribute to full and productive employment and decent work for all, in line with SDG Target 8.5. The project ensures that average salaries are paid to employees regardless of sex, age, or disability, thereby fostering inclusivity and driving economic development. Additionally, the project generates revenue from municipal waste management, including the post-combustion ash and the recovery of valuable metals and non-metals from waste. The monitoring parameters include the number of employees, disaggregated by sex, age group, and disability status, as well as average earnings and the policy for non-discrimination and equal pay for work of equal value. VVB has reviewed the project and confirms that it aligns with the objective of promoting equal and fair employment practices. The project's commitment to non-discrimination and equal pay contributes to increased productivity and economic growth, making it a significant contributor to the achievement of SDG Target 8.5 by 2030.

Sustainable cities and communities (SDG-11): The project aligns with the goal of making cities and human settlements inclusive, safe, resilient, and sustainable by contributing to SDG Target 11.6, which aims to reduce the adverse per capita environmental impact of cities. Specifically, the project focuses on improving municipal solid waste management, a key aspect of urban sustainability. By processing municipal solid waste in a controlled facility, the project enhances waste management practices and contributes to better air quality. The net inlet quantity of municipal solid waste, measured in tonnes per day, serves as the monitoring parameter to assess the project's impact. VVB has reviewed the project's approach and confirms that it effectively addresses the need for improved municipal waste management. By managing waste in a controlled facility, the project reduces the environmental impact of cities, supporting the achievement of SDG Target 11.6. The VVB concludes that the project is a significant contributor to creating more sustainable and resilient urban environments.

Responsible consumption and production (SDG-12): The project supports the goal of ensuring sustainable consumption

and production patterns by aligning with SDG Target 12.5, which aims to substantially reduce waste generation through prevention, reduction, recycling, and reuse by 2030. The project adopts material conservation and waste recycling approaches in both plant design and operations, focusing on the recovery of metal and non-metal waste from municipal solid waste (MSW). This process not only reduces waste but also promotes a circular economy by reusing recovered materials. The project's impact is monitored through the net inlet quantity of municipal solid waste and the quantity of scrap metal recovered, both measured in tonnes per day. VVB has assessed the project and concludes that its strategies for waste reduction and material recovery are effectively aligned with the objectives of SDG Target 12.5. The project's emphasis on recycling and reuse significantly contributes to reducing waste generation and fostering sustainable production patterns, making it a vital initiative in promoting circular economy practices.

Climate action (SDG-13): The project is aligned with the goal of taking urgent action to combat climate change and its impacts, specifically targeting SDG 13.2, which involves integrating climate change measures into national policies, strategies, and planning. By reducing an annual average of 57,719 tCO₂ equivalent greenhouse gas (GHG) emissions, the project achieves this through coal replacement and the prevention of open waste dumping. The generation of renewable energy-based electricity further mitigates the CO₂ emissions that would have otherwise been produced by grid-connected power plants. Since the project is fully commissioned and actively reducing emissions, it is in compliance with SDG 13. VVB has reviewed the project and concludes that it successfully integrates climate change measures, as evidenced by its substantial reduction in GHG emissions. The monitoring of GHG emission reductions (tCO₂/year) provides a clear metric for the project's effectiveness in contributing to national and global climate change mitigation efforts. The project is therefore compliant with the objectives of SDG 13.

Partnership for the goals (SDG-17): The project supports the goal of strengthening the means of implementation and revitalizing the Global Partnership for Sustainable

	Development, specifically targeting SDG 17.17, which encourages and promotes effective public, public-private, and civil society partnerships. The Waste-to-Energy (WTE) plant exemplifies this approach through its Public-Private Partnership (PPP) between M/s Antony Lara Renewable Energy Pvt. Ltd. (ALREPL) and the Pimpri Chinchwad Municipal Corporation (PCMC). This collaboration effectively combines resources and expertise from both the private sector and local government to advance sustainable waste management and renewable energy solutions. VVB confirms that the establishment of the WTE plant through this PPP aligns with the target of promoting effective partnerships. The project demonstrates a successful model of public and private sector cooperation, enhancing resource utilization and achieving shared sustainability objectives.
Additional information relevant to the project	Leakage management for AFLU project: - Not applicable to the project activity. Commercially sensitive information: - No commercially sensitive information has been excluded from the public version of the project description. VVB team conclude that the proposed project activity is eligible to register under VERRA registry and all calculations related to emission reduction are found correct. Thus accepted.

3.2 Safeguards and Stakeholder Engagement

3.2.1 Stakeholder Engagement and Consultation

3.2.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	During on-site visit interviews and based on submitted stakeholder meeting records/23/, it was confirmed that, the project stakeholders were prioritized by identifying both direct stakeholders, who have a direct interest or influence on the project, and indirect stakeholders, whose interest is more peripheral. This identification process considered the potential impacts from all facilities related to the project, including associated facilities and transport routes. As a

	result, stakeholders such as local residents, employees, local authorities, NGOs, and drivers were identified. VVB has thoroughly reviewed the stakeholder identification process and confirms that it was conducted with a comprehensive approach, considering all relevant factors associated with the project. The VVB concludes that the stakeholder prioritization effectively accounted for both direct and indirect impacts, ensuring that the identified stakeholders are accurately representative of the project's scope, as confirmed during the onsite visit.
Legal or customary tenure/access rights	The legal and customary rights to the land remain with the Pimpri Chinchwad Municipal Corporation (PCMC) throughout the project's implementation and beyond. As per the concession agreement, the project proponent is obligated to pay a license fee to PCMC for the 21-year concession period but holds no legal ownership of the land. No disputes or conflicts over land rights or resources have been identified, and the project has received full consent and support from all stakeholders. VVB has verified the land rights and confirms that the project complies with all legal requirements. The absence of disputes and the confirmed stakeholder support demonstrate that the project is fully compliant with the legal frameworks set by the landowner, PCMC.
Stakeholder diversity and changes over time	The project is located within designated area for waste management by the local government in Moshi, Pimpri Chinchwad. By promoting additional livelihood opportunities and improving sanitary conditions, the project will enhance the well-being of local communities. VVB noted no impact on stakeholder diversity and it will not change over time as it has no impact on local project area as the project abides by local laws and regulations.
Expected changes in well-being	The project effectively avoids methane emissions through the proper disposal of municipal solid waste and the generation of electricity. In the absence of this project, the baseline scenario would have involved methane emissions from landfill sites and electricity generation from grid-connected plants. By addressing these issues, the project has not only improved local sanitation and socio-economic performance but also enhanced the overall well-being of the community. VVB confirms that the project's implementation leads to significant environmental

	and socio-economic benefits, aligning with its objectives to improve local living conditions and reduce emissions.
Location of stakeholders	The community members affected by the project, both directly and indirectly, are primarily local residents living near the project site. VVB confirms that the stakeholder identification process appropriately considered these local community members. VVB noted that PP has conducted 2 step stakeholder consultation – first at the time of ESIA and second post the project commissioning to check and understand pre and post issues/grievances by local stakeholders, if any. The VVB concludes that the project has been designed to address and mitigate impacts on these stakeholders, ensuring that their concerns and interests are adequately accounted for in the project's execution.
Location of resources	The land and resources utilized for the project are owned and managed by the Pimpri Chinchwad Municipal Corporation (PCMC). All rights related to the land, including access and customary rights, are held by PCMC, which resides within the project study area. VVB confirms that the project is fully compliant with legal and customary land rights, as all relevant permissions and agreements with PCMC have been secured.

3.2.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	PP has conducted 2 stage stakeholder consultation. First stakeholder consultation was held as a part of ESIA conducted in year 2021. The other meeting was held in November, 2023. Invitations for this Local Stakeholder Consultation (LSC) were sent to the Commissioner's office on 25-October-2023, the Maharashtra Pollution Control Board on 31-October-2023, and local residents. The consultation provided ample opportunity for attendees to discuss the project, address their concerns, and offer feedback. VVB has confirmed this process through the review of the submitted invitation letters/23/.
Consultation outcome	PP held a discussion meeting with stakeholders to cover all relevant details about the project. These sessions were designed to ensure that information was shared equitably among participants, regardless of caste, religion, gender, or race, thereby minimizing any barriers to participation. Topics discussed included project design, location,

	objectives, costs, benefits, risks, duration, scope, size, impacts, and the carbon project cycle, including validation and verification processes. During these discussions, stakeholders had the opportunity to provide feedback and ask questions. The feedback received was unanimously positive, with no negative comments reported as confirmed based on feedback forms checked. The stakeholders expressed strong support for the project, and no further action is required based on the feedback. The VVB has reviewed the consultation process/23/ and confirms that it was conducted fairly and inclusively, with all stakeholder inputs being positively received and no concerns raised.
Ongoing communication	A dedicated email, phone number, and physical address for Antony Lara Renewable Energy Pvt. Ltd. were provided to all stakeholders to ensure open communication as evident based on the invitation letters checked/23/. The team will be actively present in the project area to address any queries or concerns and will conduct consultations in a culturally sensitive manner. The VVB confirms that these arrangements facilitate effective and respectful stakeholder engagement, ensuring prompt resolution of concerns and transparent management of the project's community impact.
Stakeholder input	No negative feedback was received from any local stakeholders regarding the project. All comments were overwhelmingly positive, indicating strong support for the project. Consequently, no further action is required. The VVB concludes that the positive stakeholder feedback reflects strong community support and validates the project's alignment with local interests and concerns.

3.2.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	The Pimpri Chinchwad Municipal Corporation (PCMC) retains both legal and customary rights to the land used for the project, both during and after its implementation. Under the concession agreement/17/, the project proponent is required to pay a license fee to PCMC for the land over the 21-year concession period. The project proponent does not have any legal ownership rights to the land. There are no identified disputes or conflicts related to land or resource rights, and no concerns have been raised about the project's implementation as

	during on-site visit, it was noted that the plant is in continual operation from the date of commissioning. Onsite interviews and research on the internet confirmed that the project has no ongoing litigation/issue with respect to land or resource rights. All stakeholders have expressed their consent and full support for the project. VVB has reviewed the relevant documentation and confirms that the project is in full compliance with the concession agreement and applicable legal requirements. The lack of disputes and the positive stakeholder support affirm that the project is operating as per the agreed terms and has secured all necessary approvals for successful implementation.
Outcome of FPIC discussion	Under the concession agreement with Pimpri Chinchwad Municipal Corporation (PCMC)/17/, the project proponent has the legal rights to implement the project and claim the resulting greenhouse gas (GHG) emission reductions. VVB has confirmed that the project has not involved in land encroachment or displacement of individuals as it is constructed and commissioned within the waste management premises of PCMC. VVB further confirmed that the project proponent's rights and responsibilities are fully compliant through the concession agreement/17/.

3.2.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	The project team established a structured process for receiving feedback and addressing grievances in a culturally sensitive manner, which was developed through initial consultations with stakeholders. After receiving majority approval and no further input, this procedure was finalized. VVB has reviewed the supporting documentation related to the grievance and feedback process/23/. The VVB confirms that the process was developed and finalized in accordance with culturally appropriate practices and stakeholder input. The VVB verifies that the procedure is effective for addressing concerns and grievances, ensuring that stakeholder engagement is managed transparently and inclusively.
Grievance redress procedure	A grievance mechanism is in place to address conflicts and concerns between the project proponent and local stakeholders. Stakeholders have access to the project proponent's contact information, including email and phone numbers, to facilitate open communication. VVB has

	reviewed this mechanism/23/ and confirms its effectiveness in ensuring transparent communication. The VVB verified that the provided contact details are accurate and accessible, and that the mechanism aligns with best practices for stakeholder engagement.
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3.2.1.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
During public comment period, the project has not received any comments. Even during the local stakeholder meeting held, there were no concerns raised by the local stakeholders. The potential benefits of the project activity for the local stakeholders were acknowledged. Further, during stakeholder meetings, no negative comments have been received on project activity from any of the local stakeholders consulted. As all comments were very positive about the project, no further action is required. There were no further comments raised by the stakeholders and they were totally in support for setting up of these kinds of projects in the region.	Project proponent has requested the stakeholders to contact the site in-charge any time through the email or phone or to the site address mentioned in the invitation to express their grievances in future. Also assured that a grievance register is always made available at the project site to register their complaints if any in the register and same will be addressed and resolved in the earliest possible time. The meeting list of stakeholders, Invitation to commissioner office, MPCB and photographs are provided in the Appendix.	The PP adhered to VCS Standard version 4.7 by initiating a 30-day public comment period from 21-November-2023 to 21-December-2023, enabling stakeholders and the public to provide feedback. VVB verified this compliance through the web interface and confirmed that no comments were received during this period. To facilitate ongoing communication, the project proponent has encouraged stakeholders to reach out to the site in-charge via email, phone, or the address provided in the invitation for any future grievances. A grievance register/23/ is maintained at the project site to ensure that any complaints are documented and addressed promptly. Supporting documentation, including the meeting list of stakeholders, invitations to the commissioner office and Maharashtra Pollution

		Control Board (MPCB), and relevant photographs, are provided to VVB/23/.
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3.2.2 Respect for Human Rights and Equity

3.2.2.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination and sexual harassment	The organization has established comprehensive policies/24/ to address discrimination and sexual harassment, including preventive measures and a structured grievance redressal mechanism. These policies outline the steps to be taken to prevent such activities in the workplace and provide clear procedures for reporting and resolving complaints. VVB has reviewed the relevant supporting documentation/24/, including the organization's anti-discrimination and anti-harassment policies/24/, as well as the complaint and grievance redressal mechanisms.
Management experience	The project Proponent is waste management business ¹³ and has employed staff who is well versed with the plant operation as confirmed during the on-site visit. Hence VVB accepted that the project operational team has a management experience for the project's continual operation.
Gender equity in labor and work	The project site has created job opportunities for women, with the company's HR policy ensuring equal treatment for all employees regardless of gender. VVB has reviewed the relevant supporting documentation, including the company's HR policies and employment records/25/.
Human trafficking, forced labor, and child labor	The HR policy is designed to prevent harassment in the workplace, human trafficking, forced labour, and child labour. It outlines clear measures to uphold ethical practices and protect employees' rights. VVB has reviewed the relevant supporting documentation, including the HR policy/24/ and related compliance records/24/.

3.2.2.2 Human Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Human rights	The project proponent ensures that organizational practices are aligned with human rights as per national and regional regulations. The Pimpri

¹³ <https://www.antony-waste.com/our-story/key-subsiidiaries/antony-lara-renewable-energy/>

Chinchwad Municipal Corporation (PCMC) holds the customary rights to the land and its resources/17/. The project upholds the protection of local communities' rights and customary rights holders, in line with international human rights laws and the United Nations Declaration. VVB has verified through relevant documentation that the project complies with both national¹⁴ and international human rights standards¹⁵¹⁶, effectively respecting the rights of local communities and the customary rights held by PCMC.

3.2.2.3 Indigenous Peoples and Cultural Heritage

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Preservation and protection of cultural heritage	The waste-to-energy project, located on land owned by Pimpri Chinchwad Municipal Corporation (PCMC), focuses on converting municipal solid waste into renewable energy. The project does not interact with or alter any cultural heritage sites, as it involves only waste processing and energy generation. VVB has confirmed that the project adheres to cultural heritage protection standards and does not impact local cultural heritage.

3.2.2.4 Property Rights

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Rights to territories and resources	The legal and customary rights to the land utilized for the project, both during and after its implementation, are retained by the Pimpri Chinchwad Municipal Corporation (PCMC). The project proponent is obligated to pay a license fee to PCMC for the land throughout the 21-year concession period, without holding any legal ownership rights to the land/17/. There are no disputes or conflicts related to the rights to the land or its resources. VVB has thoroughly reviewed the relevant documentation, including the concession agreement/17/, and has confirmed that the project is fully compliant with these agreements. The VVB's assessment verifies that there are no disputes or conflicts concerning land rights, ensuring that the project operates in accordance with the legal framework/17/ set by PCMC.

¹⁴ https://www.mha.gov.in/sites/default/files/Protection%20of%20HR%20Act1993_0.pdf

¹⁵ <https://www.un.org/en/about-us/universal-declaration-of-human-rights>

¹⁶ https://www.un.org/development/desa/indigenouspeoples/wp-content/uploads/sites/19/2018/11/UNDRIP_E_web.pdf

Respect for property rights	The project respects all stakeholder property rights, with the Pimpri Chinchwad Municipal Corporation (PCMC) retaining all land rights except for the carbon credits generated by the project. The Project Proponent ensures that these rights are preserved and protected as per the concession agreement/17/ The VVB verifies that the project respects stakeholder property rights/17/ and adheres to legal and contractual obligations/17/, including the proper management of carbon credits.
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3.2.2.5 Benefit Sharing

The project activity does not have benefit sharing agreement with anyone as evident from the interviews with PP representative and hence this section is not applicable.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Process used to design the benefit sharing plan	Not Applicable.
Summary of the benefit sharing plan	Not Applicable
Approval and dissemination of benefit sharing plan	Not Applicable

3.2.3 Risks to Local Stakeholders and the Environment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Risks to stakeholder participation	The assessment team confirmed the absence of identified risks for local stakeholders and the environment. This determination is prominently reflected in the stakeholder consultation documents and local stakeholder gatherings/23/. VVB team observed that the project proponent has mitigated this risk by ensuring a more extensive and comprehensive

	stakeholder interaction channel. To address any grievances or inquiries, the project proponent has developed a grievance redressal mechanism. Moreover, the project proponent is obligated by government regulations to adhere to best practices, a commitment reaffirmed during site visit interviews with both the project proponent and local stakeholders.
Working conditions	Employee productivity and well-being are affected by a wide range of elements that are included in the environment of work. These elements include workplace organization, surroundings, and physical elements. Through the project activity's rules and regulations, the VVB team verified that the working conditions were determined to be in compliance with the labor laws¹⁷ of the country. Same was confirmed during site visit interviews with employees.
Safety of women and girls	The assessment team verified that the project activity offers employment opportunities for women in both management and operational roles/25/. The project diligently tracks the number of female staff employed. Additionally, a committee has been established by the project proponent to address any complaints raised by female employees, ensuring compliance with company policy. Equal pay rights are guaranteed, and any form of discrimination in hiring or salary is strictly prohibited. These findings were corroborated during interviews with monitoring staff conducted during an onsite visit, reaffirming alignment with the company's policies.
Safety of minority and marginalized groups, including children	The assessment team confirmed that the project activity is located at existing waste disposal site identified by urban local body (ULDB) i.e. PCMC. Thus, the project does not need to show any adherence with safety of minority and marginalized groups. Additionally, the absence of child labor was noted during the visit, further affirming compliance with child protection policies¹⁸.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)	Upon verification, the assessment team found no identified risks for local stakeholders or the environment associated with the project. During the site visit, it was observed that no air, noise, or hazardous waste was generated at the project site. VVB team further confirmed that the project activity complies with the laws of the host country. Above, observations demonstrate the project's commitment to environmental and stakeholder welfare, ensuring alignment with legal requirements.

¹⁷ <https://clc.gov.in/clc/labour-law>

¹⁸ <https://labour.gov.in/childlabour/about-child-labour>

3.2.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	The project has no impact on biodiversity or ecosystems as it is located on waste disposal facility which is devoid of any bio-diversity, confirmed during an on-site visit. The land, devoid of significant flora or fauna, does not support diverse plant or animal life, so the waste-to-energy operations do not disrupt any ecosystems. The project's activities proceed without affecting local biodiversity, aligning with environmental conservation principles and avoiding ecological degradation.
Soil degradation and soil erosion	The land designated for the project is a government designated land for waste disposal site which was barren and unsuitable for agriculture, as confirmed during an on-site visit. Therefore, using this land for waste-to-energy operations aligns well with its existing condition, ensuring that no agricultural resources are affected.
Water consumption and stress	The project involves water conservation measures to ensure the plant does not consume any fresh water, as project generate electricity from waste to energy generation plant.
Usage of fertilizers	The project generates electricity through a waste-to-energy plant and does not use fertilizers.

3.2.4.1 Rare, Threatened, and Endangered species

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Species and habitat	During the onsite visit, VVB team confirmed that there were no adverse effects attributed to the project activity. This observation was further verified through discussions with end-users and other local stakeholders.

3.2.4.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
Not applicable	The assessment team verified that project is electricity generation from waste to energy plant based in Moshi, Pune, India. It was confirmed that project sites do not host any threatened plant or animal species, as validated through annual reports from the

Ministry of Environment, Forest, and Climate Change¹⁹ (MoEFCC), the nodal Ministry for implementing the Convention on Biological Diversity²⁰ (CBD) in India.

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
Not Applicable	Assessment team confirmed that project is electricity generation from waste to energy plant based in Moshi, Pune, India. During the on-site visit, the inspection team discovered that there are no invasive species present at the project site, which was also corroborated through document review and interviews with local stakeholders.

3.2.4.3 Ecosystem conversion

Item	Evidence gathering activities and evidence checked
Ecosystem conversion	During the onsite visit, the Assessment team engaged with a local stakeholder present throughout the interview, confirming the absence of endangered plant or animal species at the project location. Furthermore, no areas were identified that could disrupt valuable ecosystems.

3.3 Application of Methodology

3.3.1 Title and Reference

The PP has used type III and type I methodology for the current project as it involves methane avoidance from biomass decay and subsequent electricity generation by burning the RDF in an incinerator.

The applied methodologies are as below and VVB confirmed that the most recent versions are referred to.

Type III, Other project activities:

AMS III.E “Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment”, Version 17.0.

Type I, Renewable Energy Projects:

AMS I.D, “Grid connected renewable electricity generation”, Version 18.0.

¹⁹ <https://moef.gov.in/moef/index.html>

²⁰ <https://moef.gov.in/moef/division/environment-divisions/conservation-and-survey-cs/convention-on-biological-diversity-cbd/index.html>

The applied methodologies refers to the latest approved version of the following tools which is checked and confirmed.

Tool 03- Tool to calculate project or leakage CO2 emissions from fossil fuel combustion, Version 03.0²¹

Tool 04- Tool to calculate emissions from solid waste disposal sites, Version 08.1²²

Tool 07- Tool to calculate the emission factor for an electricity system, Version 07.0²³

Tool 12- Project and leakage emissions from transportation of freight, Version 01.1.0²⁴

Tool 21- Demonstration of additionality of small-scale project activities, Version 13.1²⁵

Tool 27- Investment analysis, Version 14.0²⁶

3.3.2 Applicability

Methodology ID	Applicability condition	Assessment and conclusion
AMS III.E, ver. 17.0	The project activity does not recover or combust methane unlike AMS-III.G. Nevertheless, the location and characteristics of the disposal site in the baseline condition shall be known, in such a way as to allow the estimation of its methane emissions.	During on-site visit, VVB noted that the project involves methane avoidance through MSW diversion from the existing landfill site and generating the electricity through controlled combustion within the boiler which is cross checked and confirmed through concession agreement/17/ and consent to operate/11/. The existing landfill site were visited. In the baseline scenario, the MSW was disposed of at these sites was confirmed by the stakeholders interviewed. The visual inspection has confirmed that they can be considered as anaerobically managed (with low porosity and waste stored was found to be moist) with volume to

²¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-03-v3.pdf>

²² <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-04-v8.1.pdf>

²³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

²⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-12-v1.1.0.pdf>

²⁵ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-21-v13.1.pdf>

²⁶ [EB122_repan01_TOOL27_ver14](#)

		surface ratio more than 1.5 thus in line with definition provided in Tool 04, version 08.1 and hence accepted. Thus, on the basis of on-site visit/interviews and consent to establish/11/, concession agreement/17/, and the commissioning certificate/14/ VVB confirms that this criterion is fulfilled.
	If the project involves combustion, gasification or mechanical/thermal treatment of partially decayed waste mined (i.e., removed) from a solid waste disposal site in addition to freshly generated waste the project proponents shall demonstrate that there is adequate capacity of the combustion, gasification or mechanical/thermal treatment facility to treat the newly generated wastes in addition to the partially decayed wastes removed from the disposal site. Alternately justifications for combusting, gasifying or mechanically/thermally treating the partially decayed wastes instead of the newly generated wastes shall be provided.	The project activity sources fresh waste from the Pimpri Chinchwad municipal corporation limits and combust the same for electricity generation as confirmed during the on-site visit and concession agreement/17/. As confirmed based on concession agreement/17/, consent to operate/11/ and on-site visit, the project does not involve combusting, gasifying or mechanically/thermally treating the partially decayed wastes. Hence this criterion is fulfilled
	If the combustion facility, the produced syngas, producer gas or RDF/SB is used for	During on-site, VVB confirmed that RDF produced is further combusted to produce

	heat and electricity generation within the project boundary, that component of the project activity may use a corresponding methodology under Type I project activities.	electricity and is sold to DISCOM based on Concession agreement/17/. The assessment team confirmed from the Concession agreement submitted that the electricity is sold to state electricity distribution company. The installed capacity is 14 MW which was confirmed from the commissioning certificate/14/. The project capacity is less than 15 MW and thus the project is a small scale project. Hence, PP has applied AMS I.D methodology which is an approved small scale methodology and hence acceptable. Hence this criterion is fulfilled.
	In case of RDF/SB production, project proponents shall provide evidence that no GHG emissions occur, other than biogenic CO₂, due to chemical reactions during the thermal treatment process for example limiting the temperature of thermal treatment to prevent the occurrence of pyrolysis and/or the stack gas analysis.	During on-site visit, VVB confirmed that RDF produced is combusted to produce electricity and no GHG other than biogenic CO₂ is generated during the process. This is confirmed based on waste characterisation report for RDF to be burned, operational parameters established in consent from PCB and boiler inspectorate certificate which has to be followed by PP. As per submitted consent from PCB, stack gas parameters were checked which has to be followed by PP as per regulatory requirement and the same confirmed that no

		other GHG is generated through the process. Hence this criterion is fulfilled.
	In case of gasification, the process shall ensure that all the syngas produced, which may contain non-CO ₂ GHG, will be combusted and not released unburned to the atmosphere. Measures to avoid physical leakage of the syngas between the gasification and combustion sites shall also be adopted.	During on-site, VVB confirmed that RDF produced is combusted to produce electricity and involves no gasification in this process as cross checked based on single line diagram/8/ also. Hence this criterion is fulfilled.
	In case of RDF/SB processing, the produced RDF/SB should not be stored in such a manner as resulting in high moisture and low aeration favouring anaerobic decay. Project proponents shall provide documentation showing that further handling and storage of the produced RDF/SB does not result in anaerobic conditions and do not lead to further absorption of moisture.	During on-site, VVB confirmed that RDF generated is with low moisture content which was confirmed based on visual inspection and test reports of the RDF submitted/33/. Further, it is not stored but immediately burned in the boiler on First In First Out (FIFO) basis as evident from discussion with PP representatives and on-site observations. The storage resulting in high moisture is not possible as designated pre-processing section in RCC bunker is an enclosure and no water is added to RDF before burning in the boiler as confirmed during on-site visit. Further, as PP has confirmed in case of shutdown, RDF is stored within the designated preprocessing section in RCC bunker as the bunker has

		enough capacity as confirmed based on DPR and on-site visit wherein development of anaerobic conditions is minimal. Hence this criterion is fulfilled.
	In case of RDF/SB processing, local regulations do not constrain the establishment of RDF/SB production plants/thermal treatment plants nor the use of RDF/SB as fuel or raw material.	VVB has checked the consent received from state pollution control board/11/ for operating the RDF processing unit. Hence this criterion is fulfilled.
	During the mechanical/thermal treatment to produce RDF/SB no chemical or other additives shall be used.	VVB during the on-site visit noted that the MSW after pre-treatment is processed through mechanical means which results in RDF production ready for burning in the boiler. This was confirmed during on-site visit and confirmed based on consent by PCB/11/ and single line diagram/8/ which does not mentioned provision for chemical/additives to produce RDF. Hence this criterion is fulfilled.
	In case residual waste from controlled combustion, gasification or mechanical/thermal is stored under anaerobic conditions and/or delivered to a landfill, emission from the residual waste shall to be taken into account using the first order	During on-site visit, VVB has checked and confirmed that bottom ash and fly ash is not stored under anaerobic condition but is stored in two designated silos/32/. The generated bottom ash and fly ash are collected in tractor trolleys. VVB during on-site visit, checked the

	decay model (FOD) described in AMS-III E	records of bottom ash and fly ash generation which is handed over to the brick manufacturers on daily basis which is confirmed based on ash transfer records/32/. Thus, VVB ensured that emissions from the residual waste need not to be considered.
AMS I.D, ver. 18.0	This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or Involve a replacement of (an) existing plant(s).	This is a greenfield project established to treat MSW and generate electricity from the same. This is confirmed based on consent to establish/operate from state PCB/11/ and Concession Agreement/17/. All these documents confirmed that there was no existing plant at the project site prior to the current project which treated MSW and generated electricity. Hence this criterion is fulfilled.
	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the	The mentioned criterion is not applicable as it is a grid connected waste to energy project activity and not a hydro power project activity. The same was verified by the assessment team on the basis of the commissioning certificate/14/ submitted by the PP.

	power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m²; The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m².	
	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	This is a greenfield project established to treat MSW and generate electricity from the same. The assessment team on the basis of the commissioning certificate/14/ and onsite audit confirms that the unit do not co-fires fossil fuel. Hence this criterion is not applicable.
	Combined heat and power (co-generation) systems are not eligible under this category.	This is a greenfield project established to treat MSW and generate electricity from the same. The assessment team on the basis of the commissioning certificate and onsite audit confirms that no co-generation system is involved. Hence this criterion is not applicable.
	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable	This is a greenfield project established to treat MSW and generate electricity from the same. The assessment team on the basis of the

	power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	commissioning certificate/14/ and onsite audit confirms that it is installation of new grid connected waste to energy plant and no capacity addition is involved. Hence this criterion is not applicable.
	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	This is a greenfield project established to treat MSW and generate electricity from the same. The assessment team on the basis of the commissioning certificate and onsite audit confirms that it is installation of new grid connected waste to energy plant and no retrofit, rehabilitation is involved. Hence this criterion is not applicable.
	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed	This is a greenfield project established to treat MSW and generate electricity from the same. This is confirmed based on consent to establish/operate from state PCB/11/ and concession Agreement/17/. The methane avoidance part is claimed under AMS III.E as demonstrated and assessed earlier. VVB noted that as per concession agreement/17/, fresh waste is directly deposited at the project site by municipal corporation and

	under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored.	no methane recovery is possible from waste as deposited waste after processing is directly burned in the boiler. Hence this criterion is fulfilled.
	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	The assessment team on the basis of onsite audit, the commissioning certificate/14/ and Consent to establish by the Pollution control board/11/ confirms that the project activity involves treatment of municipal waste generated from Pimpri Chinchwad municipal corporation limits to produce RDF. Thus, the project activity do not involves biomass. Hence this criterion is not applicable.
Tool 03 Title: Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion, Version 03.0	This tool provides procedures to calculate project and/or leakage CO₂ emissions from the combustion of fossil fuels. It can be used in cases where CO₂ emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties. Methodologies using this tool should specify	As observed during on-site visit, PP uses diesel for waste handling vehicles, waste loaders into bunker and for DG set used as a backup in case of emergency. Hence this criterion is fulfilled.

	to which combustion process j this tool is being applied.	
Tool 04, Title: Tool to calculate emissions from solid waste disposal sites, Version 08.1	The tool can be used to determine emissions for the following types of applications Application A: The CDM project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g., “ACM0001: Flaring or use of landfill gas”). The methane is generated from waste disposed in the past, including prior to the start of the CDM project activity. In these cases, the tool is only applied for an ex-ante estimation of emissions in the project design document (CDM-PDD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e.g., measuring the amount of methane captured from the SWDS); Application B: The CDM project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or	VVB noted during the on-site visit that the project diverts fresh MSW from reaching the landfill site and it is combusted after mechanical. Treatment to generate electricity in line with Concession Agreement between Pimpri Chinchwad Municipal Corporation and Antony Lara Renewable Energy Pvt. Ltd (ALREPL). Hence application B is appropriate scenario.

	anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex-ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	
Tool 07 Title: Tool to calculate the emission factor for an electricity system, Version 07.0	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	This is a greenfield project established to treat MSW and generate electricity from the same which is fed into the national electricity grid. This is confirmed based on consent to establish/operate from state PCB/11/ and concession Agreement/17/ along with on-site visit. All these documents confirmed that there was no existing plant at the project site prior to the current project which treated MSW and generated electricity.
	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project	As confirmed during on-site visit, the project activity is selling the generated electricity to the local electricity grid i. e. the project is grid connected waste to energy project. Hence this condition is applicable to the project activity.

	proponents, i.e., option II a and option IIb. If option II a is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	The assessment team checked and confirmed that PP has calculated combined margin grid emission factor as per the CO₂ emission factor based on CEA database published by Central Electricity Authority (CEA)/34/ which for the calculation of emission factor has only considered grid connected plants. Thus, the criterion is satisfied.
Tool 12 Title: Project and leakage emissions from transportation of freight, Version 01.1.0	This tool is applicable to project activities which involve freight transportation by road and where transportation is not the main project activity. This tool is not applicable to project activities where transportation is the main source of greenhouse gases emissions. This tool does not provide procedures to estimate baseline emissions from road transportation of freight. The tool only provides to determine CO₂ emissions.	VVB during on-site visit noted that fresh MSW is transported from the city to the project site, the same is verified by the assessment team on the basis of the Concession agreement/17/ submitted by the PP and generated ash is taken to brick manufacturers. This involves freight transportation by road and PP will calculate project emissions due to freight transportation hence this criterion is fulfilled.

	CH ₄ and N ₂ O emissions are excluded for simplification as they are small compared to CO ₂ emissions.	
	In addition, the tool is applicable for the determination of project or leakage emissions from freight transportation by rail in project activities where transportation is not the main project activity.	VVB during on-site visit noted that fresh MSW is transported from the city to the project site and generated ash is taken to brick manufacturers. This involves freight transportation by road and not by rail and also transportation is not the main project activity as per the criteria. hence this criterion is not applicable.
Tool 21 Title: Demonstration of additionality of small-scale project activities, Version 13.1	The use of the methodological tool “Demonstration of additionality of small-scale project activities” is not mandatory for project proponents when proposing new methodologies. Project proponents and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	PP is not proposing new methodologies and using existing CDM approved methodology and hence this tool is applicable.
	Project proponents and coordinating/managing entities may also apply “TOOL19: Demonstration of additionality of microscale project activities” as applicable.	The project activity is not a micro-scale project and hence tool 21 is applicable and not Tool 19.

Tool 27 Title: Investment analysis, Version 14.0	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	PP is applying “non-binding best practice examples to demonstrate additionality for SSC project activities” and hence this Tool 27 is required for investment analysis.
	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	VVB has checked and confirmed that the applied methodologies do not contain requirements for investment analysis and hence applicability of this tool is found appropriate.

3.3.3 Project Boundary

PP has defined the project boundary in accordance with both the applied methodologies:

According to paragraph 21 of AMS-III.E, version 17.0, the project boundary are the physical, geographical sites:

- Where the solid waste would have been disposed or is already deposited and the avoided methane emission occurs in absence of the project activity;
- Where the treatment of biomass through controlled combustion, gasification or mechanical/thermal treatment takes place;

- (c) Where the final residues of the combustion process will be deposited (this parcel is only relevant to controlled combustion activities);
- (d) And in the itineraries between them, where the transportation of wastes and combustion residues and/or residues of gasification and mechanical/thermal treatment process occurs.

AS per applicable methodology PP shall consider existing waste disposal landfill site, the project site where combustion of waste is taking place, ash disposal location and itineraries between them.

VVB noted that, in the project boundary, PP has considered, existing solid waste disposal site (SWDS) where the solid waste would have been disposed and the avoided methane emission occurs in absence of the project activity and the final residues of the combustion process will be deposited;

The current project activity which involves the treatment of waste through controlled combustion; And in the itineraries between existing SWDS and project activity for the transportation of wastes and combustion residues. VVB noted that, all these are clearly demonstrated and considered in the project boundary and hence acceptable,

This is clearly demonstrated in the below diagram by PP which is acceptable.

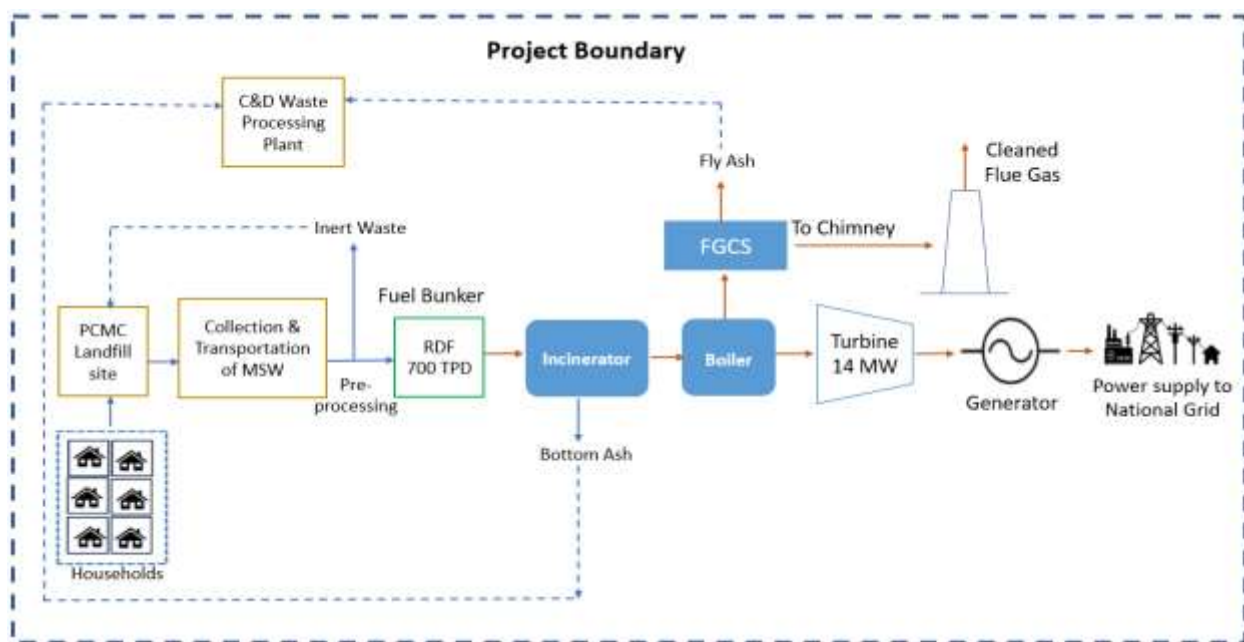


Figure 1: project boundary of the project activity

According to paragraph 18 of AMS-I. D, version 18, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the CDM project power plant is connected to.

Thus, in line with the applicable methodology PP has considered project power unit and existing national electricity grid to which this plant is connected to. The project activity involves incineration of RDF prepared from MSW and generated electricity is exported to the national grid. VVB noted that, all these are clearly demonstrated and considered in the project boundary and hence acceptable.

The project boundary as depicted by PP is as below which was checked to confirm that each GHG source, sink and reservoir is considered in accordance with the applied methodologies.

Source		Gas	Included?	Assessment and conclusion
Baseline	Emissions from decomposition of waste at the SWDS	CO ₂	No	CO ₂ emissions from the decomposition of fresh waste are not accounted.
		CH ₄	Yes	The major source of emissions in the baseline.
		N ₂ O	No	N ₂ O emissions are small compared to CH ₄ emissions from landfills. Exclusion of this gas is conservative
	Emissions from electricity generation	CO ₂	Yes	Major source. Electricity generation emission is included in electricity grid-based emissions.
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
Project	Emissions from on-site fossil fuel consumption due to the project activity	CO ₂	Yes	CO ₂ emission from fossil fuel consumption for the operation of the project activity.
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
	Emissions from on-site electricity use	CO ₂	Yes	Electricity consumed for the operation of the project activity will be supplied from the Grid.
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
	Emission from the waste treatment processes	CO ₂	Yes	CO ₂ emission generated from combustion of RDF.
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
	Emissions through	CO ₂	Yes	CO ₂ emission generated from incremental transportation.
		CH ₄	No	Excluded for simplification. This is conservative.

Source	Gas	Included?	Assessment and conclusion
incremental transportation	N ₂ O	No	Excluded for simplification. This is conservative.

3.3.4 Baseline Scenario

As per para. 28 of AMS-III.E, the baseline scenario for the project activity is the situation where, in the absence of the project activity, organic waste matter is left to decay within the project boundary and methane is emitted to the atmosphere. The yearly baseline emissions are the amount of methane that would have been emitted from the decay of the cumulative quantity of the waste diverted or removed from the disposal site, to date, by the project activity, calculated as the methane generation potential using the “Tool to determine methane emissions avoided from disposal of waste at a solid waste disposal site”.

VVB during on-site visit, through interviews with the local stakeholders and through document review such as concession agreement which mentions diversion of fresh MSW from existing dump site for the project confirmed that prior to the project implementation, MSW in Pimpri Chinchwad was sent to landfill and used to get decayed in anaerobic condition without any methane recovery system for flaring or energy recovery purpose. Thus, this anaerobic condition will release methane emissions into the atmosphere.

Thus, VVB accepts the baseline scenario as per AMS III.E is correctly established and deemed reasonable.

As per para. 19 of AMS-I. D, the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.²⁷

As confirmed based on consent to establish/operate/11/ and concession agreement/17./, the project is a greenfield project generating and feeding the same in the electricity grid. Prior to the implementation of the project activity, same amount of electricity would have been generated through fossil fuel fired grid connected energy plants.

Thus, VVB accepts the baseline scenario as per AMS I.D is correctly established and deemed reasonable.

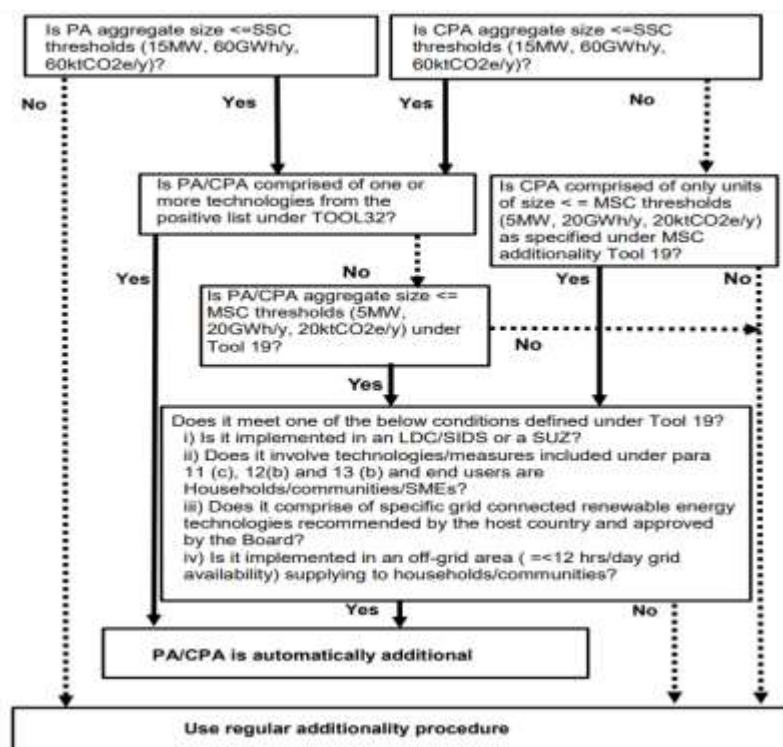
VVB confirms that PP has submitted the required documentary evidence i.e. concession agreement used in determining the baseline scenario which are relevant, are correctly quoted and interpreted in the project description.

In the VCS PD, PP has considered all the relevant national and /or sectoral policies and circumstances and have listed which were checked and confirmed.

VVB confirms that the procedures for identifying the baseline scenario have been correctly followed and the identified scenario reasonably represents what would have occurred in the absence of the project and identified baseline scenario is justified.

²⁷ https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Asia/India_Asia_RE_SP.pdf

3.3.5 Additionality



PP has followed the procedure mentioned in above process diagram:

Step 1: Is PA aggregate size <= SSC thresholds (15 MW, 60 GWh/y, 60 ktCO_{2e}/y)

The project activity's capacity is below 15 MW, thus meeting the threshold for small-scale CDM projects. The VVB team has not only confirmed this through the Concession Agreement and onsite visit but has also conducted a thorough assessment, verifying that the project adheres to all applicable eligibility criteria and regulatory requirements for small-scale projects. This ensures compliance with the necessary standards and confirms that the project qualifies within the SSC thresholds.

Step 2: Is PA/CPA comprised of one or more technologies from the positive list under TOOL 32

This project involves waste-to-energy-based grid-connected power generation. According to the latest version of Tool 32: "Positive List of Technologies" (version 04), this specific technology is not included in the positive list. The VVB team has thoroughly assessed the project and verified that, despite its exclusion from the positive list, the project complies with all other relevant requirements and standards. The team has further confirmed that the project meets the necessary criteria for registration under the applicable regulatory framework.

Step 3: Is PA/CPA Aggregate size <= MSC thresholds (5 MW, 20 GWh/Y, 20 ktCO_{2e}/y) under Tool19?

The project has an installed capacity of 14 MW, which exceeds the threshold for micro-scale activities. The VVB team has conducted a detailed assessment, confirming that the project does not qualify as a micro activity. Additionally, the VVB has verified that the project's scale and technical parameters align with the applicable regulatory frameworks for small-scale activities, ensuring that all eligibility criteria are met.

Step 4: Does it meet one of the below conditions defined under Tool 19?

1. Is it implemented in an LDC/SIDC or a SUZ ?

The project is not implemented in a Least Developed Country (LDC), Small Island Developing Country (SIDC), or a Special Underdeveloped Zone (SUZ). The VVB team has thoroughly assessed the project's location and confirmed that it does not meet the geographical criteria for these categories.

2. Does it involve technologies/measures included under para 11 (c), 12(b), and 13 (b), and are the end users Households/communities/SMEs?

The project does not involve the specific technologies or measures listed under paragraphs 11(c), 12(b), and 13(b) of the relevant guidelines. Additionally, the VVB has verified that the end users of the project are not exclusively households, communities, or SMEs, confirming that this condition does not apply.

3. Does it comprise specific grid-connected renewable energy technologies recommended by the host country and approved by the Board?

The project does not utilize any grid-connected renewable energy technologies specifically recommended by the host country and approved by the Board. After a comprehensive review, the VVB confirmed that this requirement is not relevant to the project.

4. Is it implemented in an off-grid area (≤ 12 hrs/day grid availability) supplying to households/communities?

The project is not implemented in an off-grid area with limited grid availability (≤ 12 hours per day) and does not supply energy directly to households or communities. The VVB team has verified this through a detailed assessment of the project's operational area and grid connection status.

In conclusion, based on the VVB's detailed assessment, the aforementioned conditionalities do not apply to this project and hence the regular additionality approach needs to be followed as stated below:

As per Methodological Tool 21, Tool for the demonstration of Additionality of Small-scale Project Activities (Version 13.1, EB 105, Annex 0418), to establish the project additionality, it has to be shown that the project activity would not have occurred anyway due to at least one of the following barriers:

- **Investment barrier:** a financially more viable alternative to the project activity would have led to higher emissions.

- **Technological barrier:** a less technologically advanced alternative to the project activity involves lower risks due to the performance uncertainty or low market share of the new technology adopted for the project activity and so would have led to higher emissions.
- **Barrier due to prevailing practice:** prevailing practice or existing regulatory or policy requirements would have led to implementation of a technology with higher emissions.
- **Other barriers:** without the project activity, for another specific reason identified by the project proponent, such as institutional barriers or limited information, managerial resources, organizational capacity, financial resources, or capacity to absorb new technologies, emission would have been higher.

The project proponent has selected the Investment barrier to demonstrate in a conservative and transparent manner that the project activity is financially unattractive. In line with the guidelines stipulated under Annex 34 of EB 358 (“Non-binding best practice examples to demonstrate additionality for SSC project activities”), a benchmark analysis has been used for the analysis and Equity IRR has been chosen as the financial indicator for the demonstration of the additionality.

Selection of Appropriate Benchmark:

PP has used the most recent version of “TOOL27: Investment analysis” to define appropriate benchmark. As the project activity generates power utilizing waste to energy, Group 1 as per para 5a of Appendix of EB 122, Annex 1 has been identified as a suitable category.

As per appendix of “TOOL27: Investment analysis, version 8.0”, which was available at the time of investment decision (i. e. 11-January-2018), the expected return of equity in real terms for Energy Industries (Group 1) in India = 10.73%.

TOOL 27: “Investment Analysis”, version 14.0 is the latest version available. The default value mentioned in version 14.0 is 9.13% for group 1 projects in India is used which is appropriate and more conservative for benchmark calculation and the project owner has considered the same tool version for default value of return on equity for the respective project.

The investment analysis has been carried out in Nominal terms. Accordingly, default value as given in table under the Appendix, EB 122, Annex 1 has been adjusted by adding suitable forecasted inflation rate.

The inflation forecast has been considered from the RBI²⁸ (Reserve Bank Of India) based on the investment decision date (11.01.2018) for 10 years is =4.40%

The benchmark has been computed in the following manner:

Default Value Benchmark:

²⁸ <https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=18046>

The cost of equity is determined by selecting the values provided in the table of the Appendix, i.e., Default values for cost of equity (expected return on equity) in the Tool27: Investment analysis, Version 14.0.

The Required return on equity (benchmark) was computed in the following manner:

$$\text{Nominal Benchmark}^{29} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where:

Default value for Real Benchmark = 9.13% (as per Appendix of EB 122, Annex 1) Inflation Rate forecast by RBI³⁰.

The Inflation rate forecast of the host country; India is published by the RBI (Reserve Bank of India). As per the Tool27: Investment analysis, version 14.0, the average forecasted inflation rate has considered the next 10 years average inflation rate.

Benchmark estimation:

The Cost of Equity has been considered using the “Tool27: Investment analysis” version 14.0 as per the latest available value.

Table under Appendix in EB 122, Annex 1 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in India= 9.13%

Inflation Forecast	Benchmark
4.40%	13.93%

Hence a benchmark of 13.93 % has been selected for the project activity which is found appropriate.

Initially, the consortium M/S. Antony Lara Enviro Solutions Pvt Ltd and M/S. AG Enviro Infra Projects Pvt Ltd has participated in the Request for Proposal (RfP) issued by Pimpri Chinchwad Municipal Corporation (PCMC) dated 25-September-2017.

The project proponent has participated in the bidding of Request for Proposal (RfP) issued by Pimpri Chinchwad Municipal Corporation dated 25-September-2017/19/. The project proponent had submitted the Detailed Project Report (DPR) (prepared by third party) with key assumptions and financial parameters to PCMC along with the bid submission in line with the request for proposal. At this time when the board has started evaluating the input assumptions and financial viability of the project activity to participate in the bidding and further to invest in the project activity and the bid was submitted on 11-

²⁹ https://cdm.unfccc.int/Reference/Guidclarif/ssc/methSSC_guid15_v01.pdf

³⁰ <https://www.rbi.org.in/Scripts/PublicationsView.aspx?id=18046>

January-2018 which is considered as the investment decision date. The letter of award for the implementation of the project activity was received on 28-February-2018/19/. Input values for the investment analysis are sourced from the DPR which is the first and foremost document available at the time when the project proponent started evaluating the project activity for the bidding and investment. Hence, all the input parameters used in the DPR are available at the time of decision making.

Subsequently, in line with the 4.2.3 (f) para of RfP which states “the members of a Consortium shall form an appropriate SPV to execute the Project, if awarded to the Consortium”, the Antony Lara Renewable Energy Private Limited was founded in 24-July-2018 to execute the project. Same was checked by the VVB during the on-site visit from the site personnel and was found to be accurate.

Chronology of events of the project activity is provided below:

S. No	Activity	Date of Activity
1.	Request for proposal issued by Pimpri Chinchwad Municipal Corporation (PCMC)	25-September-2017
2.	Completion of DPR by ESBEE Power Solutions Pvt Ltd.	December 2017
3.	Bid submission to PCMC – Investment decision date	11-January-2018
4.	Letter of award issued by PCMC to consortium M/S. Antony Lara Enviro Solutions Pvt Ltd and M/S. AG Enviro Infra Projects Pvt Ltd	28-February-2018
5.	Formation of the SPV for project execution - Antony Lara Renewable Energy Private Limited (ALREPL)	24-July-2018
6.	Signing of Concession Agreement between PCMC and ALREPL	06-September-2018
7.	Signing of Conversion charges Agreement between PCMC and ALREPL	26-June-2020
8.	Signing of EPC Contract between ALREPL and Hitachi Zosen India Private Limited and ISGEC HEAVY Engineering Limited	06-July-2020
	Commissioning Date of the project activity	07-October-2023

The chronology outlined above has been verified as accurate based on the bidding process. The VVB team has thoroughly reviewed and assessed the relevant information provided in the supporting documents and confirmed that the details are consistent with the established bidding timeline and

procedures. All necessary documentation supports this confirmation, ensuring the integrity of the process.

Calculation and comparison of financial indicators

The period considered for Post Tax Equity IRR calculations is 21 years, which corresponds to the concession agreement period.

Depreciation, and other non-cash items related to the project activity, which have been deducted in estimating gross profits on which tax is calculated, is added back to net profits for the purpose of calculating the financial indicator.

Appropriateness for consideration of Input Parameters:

1. VVB noted that as per the concession agreement/17/, the tipping fee (i.e., revenue from processing of the MSW) is INR 504 / tonne of waste which is being paid by PCMC to the PP and the same has been considered for investment analysis.
2. The on-site visit and on-site records/24/ have confirmed that the bottom ash and fly ash is sent to Construction and Demolition waste processing plant for free of cost. Thus, PP has not considered the revenue from bottom ash and fly ash for the investment analysis.
3. As per para 3 of EB48 Guidelines, Annex 11, the plant load factor shall be defined ex-ante in the CDM-PDD according to one of the following three options,
 - a. The plant load factor provided to banks and/or equity financiers while applying the project activity for project financing, or to the government while applying the project activity for implementation approval;
 - b. The plant load factor determined by a third party contracted by the project proponents (e.g. an engineering company);

The PLF value considered for investment analysis is 80% sourced from DPR dated December 2017 prepared by ESBEE Power Solutions Pvt. Ltd./16/ which is a third-party engineering company and hence the DPR is in line with the requirements of para 3(b) of “Guidelines for the reporting and validation of plant load factors”. Thus, the PLF considered for the investment analysis is found to be in compliance with EB48 Annex 11³¹ as the DPR was prepared by third party engineering company.

4. In India, generally the power purchase agreement (PPA) is signed with fixed tariff rate over the lifetime of the project activity. Hence a fixed tariff rate with no escalation has been considered for the investment analysis. For the project activity, the PCMC has signed a conversion charges

³¹ https://cdm.unfccc.int/EB/048/eb48_repan11.pdf

agreement to utilize the electricity generated by the project through the grid for a fixed tariff of 5 INR/kWh.

5. The O&M cost has been sourced from DPR available at the time of investment decision considering 3% of escalation. The O&M cost is comprised of the spares, employee cost, administrative and general expense, consumables, repairs and maintenance and insurance expenses, etc.

Input values considered for the IRR calculations are given below.

Particulars	Value	Unit	Source/Remarks
Quantity of MSW treated	700	TPD	DPR dated December 2017 (Page No. 69)
Capacity of the project	14	MW	DPR dated December 2017 (Page No. 127)
Operating days	330	Days	DPR dated December 2017 (Page No. 127)
PLF	80	%	DPR dated December 2017 (Page No. 127)
Annual Net generation	88.704	GWh/year	Calculated
Auxiliary Electricity Consumption	12.50	%	DPR dated December 2017 (Page No. 127)
Net Electricity Generation	77.62	GWh/year	Calculated
Project cost	2500	INR Million	DPR dated December 2017 (Page No. 127)
Debt	70	%	DPR dated December 2017 (Page No. 127)
Equity	30	%	DPR dated December 2017 (Page No. 127)
Debt	1,750.00	INR Million	Calculated
Equity	750.00	INR Million	Calculated
Grant	500.00	INR Million	RFP dated 25-September-2017
Interest rate	10.50	%	DPR dated December 2017 (Page No. 127)
Debt Repayment tenure	14	years	DPR dated December 2017 (Page No. 127)
Moratorium	1	year	DPR dated December 2017 (Page No. 127)
Operation and Maintenance (Spares, employee cost, administrative and general expense, statutory charges, repairs, maintenance and expenses)	210.0	INR Million	DPR dated December 2017 (Page No. 127)
Annual Escalation	3.00	%	DPR dated December 2017 (Page No. 127)
Service tax on O&M services	18.00	%	<u>Heading 9954 or 9983 or 9987</u>

Insurance cost (0.75% of project cost)	18.75	INR Million	DPR dated December 2017 (Page No. 127)
Tariff	5	INR/kWh	DPR dated December 2017 (Page No. 127)
Tipping Fee	504	INR/tonne	DPR dated December 2017 (Page No. 127)
Interest on working capital	11.66	%	https://www.cercind.gov.in/2017/orders/05.pdf
Depreciation Rate (Book)	4.76	%	Calculated
Project cost	2,500.00	INR Million	DPR dated December 2017 (Page No. 127)
Land cost	0	INR Million	DPR dated December 2017 (Page No. 127)
Gross Depreciable Value	2,500.00	INR Million	Calculated
Salvage Value (@ 10%)	250.00	INR Million	Calculated
Net Depreciable Value	2,250.00	INR Million	Calculated
Residual value	250.00	INR Million	Calculated
IT Depreciation Rate	7.84	%	http://www.incometaxindia.gov.in/charts%20%20tables/depreciation%20rates.htm
Income tax rate	34.27	%	Calculated from the source given below table (Income tax rate+ Income tax rate*Surcharge) + (Income tax rate+ Income tax rate*Surcharge)*Health & Education cess (%)
MAT rate	21.13	%	Calculated from the source given below table (MAT+ MAT*Surcharge) + (MAT+ MAT*Surcharge)*Health & Education cess (%)
Salvage Value	10	%	DPR dated December 2017 (Page No. 127)
Concession period	21.00	Year	DPR dated December 2017 (Page No. 19)

Financial Year	AY 2018 - 19	Source	
Income tax rate (%)	30.00	Indian IT Act for AY 2018-19	https://incometaxindia.gov.in/_layouts/15/dit/Pages/viewer.aspx?path=/Charts%20%20Tables/Tax-rates-for-last-10-years/Companies.html&IsDlg=0
MAT (%)	18.50	Indian IT Act for AY 2018-20	https://incometaxindia.gov.in/_layouts/15/dit/pages/viewer.aspx?grp=act&cname=CMSID&cval=102120000000073619&searchFilter=&k=&IsDlg=0

Surcharge (%)	12.00	Indian IT Act for AY 2018-21	https://incometaxindia.gov.in/layouts/15/dit/Pages/viewer.aspx?path=/Charts%20%20Tables/Tax-rates-for-last-10-years/Companies.html&IsDlg=0
Health & Education cess (%)	2.00	Indian IT Act for AY 2018-22	https://incometaxindia.gov.in/layouts/15/dit/Pages/viewer.aspx?path=/Charts%20%20Tables/Tax-rates-for-last-10-years/Companies.html&IsDlg=0
Final tax rate			
Income tax rate (%)	34.27%	%	Calculated
MAT (%)	21.13%	%	Calculated

Post Tax Equity IRR for the project activity against the benchmark values are shown in table below. Thus, it is evident that the project is not financially attractive as the equity IRR is below the benchmark value.

Post tax Equity IRR	
Project post-tax equity IRR	1.64%
Benchmark Value	13.93%

Sensitivity Analysis:

The robustness of the conclusion drawn above, namely that the project is not financially attractive, has been tested by subjecting critical assumptions to reasonable variation. As required by EB 122, Annex 1, only variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenues should be subjected to reasonable variation. PP has identified the total revenue from the project activity is dependent on the Tariff, Plant Load Factor, Project Cost, and O&M Costs and debt constitute more than 20% of the project costs. These factors have been subjected to a 10% variation on either side and the results of the sensitivity analysis indicate that even after applying such variation the EIRR does not cross the benchmark.

Variation %	-10%	Normal	10%	Variation required to reach benchmark	Value required to reach benchmark
Tariff (Rs/KWh)	-1.71%	1.64%	4.84%	39.14%	6.957
PLF (%)	-1.71%	1.64%	4.84%	39.14%	111.31%
Project Cost (INR Million)	3.73%	1.64%	-0.09%	-40.10%	1,497.50
O&M Cost (INRV Million)	4.41%	1.64%	-1.40%	-49.40%	106.26
Debt (INR Million)	3.29%	1.64%	0.14%	-61.20%	679

- a) **Tariff:** The tariff rate of 5.00 INR/kWh used for the investment analysis has been sourced from the Detailed Project Report (DPR). The VVB team has reviewed this data and confirmed that the post-

tax equity IRR would exceed the benchmark value only if the tariff varied by 39.14%. However, the actual tariff specified in the signed concession agreement is consistent with the estimated tariff. Based on the VVB's thorough assessment, an increase in the tariff is highly unlikely, and therefore, the benchmark IRR will not be breached.

- b) **PLF:** The Plant Load Factor (PLF) value of 80% used for the investment analysis has been sourced from the Detailed Project Report (DPR) and complies with EB48 Annex 11, as the DPR was prepared by a third-party engineering firm. The VVB team has reviewed this and confirmed that the post-tax equity IRR would breach the benchmark value only if the PLF varied by 39.14%. However, based on the VVB's assessment, it is highly unlikely that the PLF will increase to breach the benchmark, as the normative PLF for the MSW project is lower than the estimated PLF.
- c) **Project Cost:** The project cost of 2,500 million INR, used for the investment analysis, is sourced from the Detailed Project Report (DPR). A cost reduction of -40.10% would be required for the post-tax equity IRR to breach the benchmark. However, as the project has already been commissioned, such a variation is not feasible. The VVB has reviewed and confirmed that the actual incurred cost is 2,338.73 million INR and confirmed that, at this cost, the post-tax equity IRR remains below the benchmark. Therefore, a cost reduction of 40.10% is highly unlikely, ensuring that the benchmark will not be breached.
- d) **O&M Costs:** The Operation and Maintenance (O&M) cost of 210 million INR used for the investment analysis is sourced from the Detailed Project Report (DPR). A reduction of -49.40% in O&M costs would be required for the post-tax equity IRR to breach the benchmark, which is not feasible. The VVB has reviewed the O & M contract to confirm the actual O&M cost i. e. 330 million INR per year and confirmed that, even at this level, the post-tax equity IRR has not breached the benchmark. Therefore, a reduction of 49.40% in O&M costs is highly unlikely, ensuring that the benchmark remains intact.

Based on sensitivity analysis it can be concluded that the project activity is additional with reasonable variation in values and is not likely to breach the benchmark value. The occurrence of these events is unlikely.

Conclusion:

As described above, the project fulfils all necessary requirements of additionality specified in the Tool 21- Demonstration of Additionality of Small-scale Project Activities, version 13.1. Hence, the project is deemed additional.

3.3.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

PP has demonstrated quantification of GHG emission reductions and removals in section 4 of VCS PD which was checked and the VVB assessment is presented below. The calculations provided for emission reductions below were checked based on sourced documents i. e. ER sheet, detailed project report,

commissioning certificate, CEA database (<https://cea.nic.in/cdm-co2-baseline-database/?lang=en>) and other documents submitted by PP.

Baseline emissions:

Baseline Emissions from Solid Waste Disposal Sites:

As per para 17, equation (1) of tool 04, version 08.1, the baseline emissions for an year is calculated using the following formula:

$$\left. \begin{matrix} BE_{CH_4,SWDS,y} \\ PE_{CH_4,SWDS,y} \\ LE_{CH_4,SWDS,y} \end{matrix} \right\} = \varphi_y \times (1 - f_y) \times GWP_{CH_4} \times (1 - OX) \times \frac{16}{12} \times F \times DOC_{f,y} \\ \times MCF_y \times \sum_{x=1}^y \sum_j (W_{j,x} \times DOC_j \times e^{-k_j \times (y-x)} \times (1 - e^{-k_j}))$$

Where,

$BE_{CH_4,SWDS,y}$ = Baseline, project or leakage methane emissions occurring in year y generated
 $PE_{CH_4,SWDS,y}$ from waste disposal at a SWDS during a time period ending in year y (t CO₂e/yr)
 $LE_{CH_4,SWDS,y}$

x = Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)

y = Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)

DOC_{f,y} = Fraction of degradable organic carbon (DOC) that decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)

W_{j,x} = Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)

φ_y = Model correction factor to account for model uncertainties for year y

f_y = Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y

GWP_{CH₄} = Global Warming Potential of methane

OX = Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)

F = Fraction of methane in the SWDS gas (volume fraction)

MCF_y = Methane correction factor for year y

DOC_j = Fraction of degradable organic carbon in the waste type j (weight fraction)

- k = Decay rate for the waste type j (1 / yr)
- j = Type of residual waste or types of waste in the MSW

As per para 27, equation (5) of tool 04, version 08.1, the amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x is calculated as follows:

$$W_{j,x} = W_x \times p_{j,x}$$

Where,

- $W_{j,x}$ = Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)
- W_x = Total amount of solid waste disposed or prevented from disposal in the SWDS in year x (t)
- $p_{j,x}$ = Average fraction of the waste type j in the waste in year x (weight fraction)
- j = Type of residual waste or types of waste in the MSW
- x = Years in the time period for which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)

As per para 28, equation (7) of tool 04, version 08.1, the fraction of the waste type j in the waste for the year x or month i are calculated as follows:

$$p_{j,x} = \frac{\sum_{n=1}^{Z_x} p_{n,j,x}}{Z_x}$$

Where,

- $p_{j,x}$ = Average fraction of the waste type j in the waste in year x (weight fraction)
- Z_x = Number of samples collected during the year x
- $p_{n,j,x}$ = Fraction of the waste type j in the sample n collected during the year x (weight fraction)
- n = Samples collected in year x

Hence, PP has calculated the baseline emissions from Solid Waste Disposal Sites as follows in the ER sheet:

Φ_y	Model Correction Factor	0.85
1 - f_y	Fraction of methane captured	1.0
GWP_{CH_4}	Global Warming Potential of CH ₄	28
1 - OX	Oxidation factor	0.9
F	Fraction of CH ₄	0.5
$DOC_{f,y}$	Fraction of DOC decomposed	0.5
MCF_y	Methane Correction Factor	1

Quantity of RDF processed per day = 700 TPD
 Operating days in year x = 330 days
 Operating hours in day = 24 hours
 Quantity of RDF processed per year (W_x) = 231,000 Tonnes/year

Now for W_{j,x} as mentioned earlier,

$$W_{j,x} = W_x \times p_{j,x}$$

Composition	P _{j,x}	W _{j,x}	DOC _j	K _j	W _{j,x}
j	%	TPY (Wet)	%	1/y	TPD (Wet)
Food	0.00%	0	15%	0.085	0.00
Paper	19.71%	45,530	40%	0.045	137.97
Wood including Coconut Shell	2.20%	5,082	43%	0.025	15.40
Green/Mix	46.39%	107,161	20%	0.065	324.73
Plastic	8.70%	20,097	0%	0.000	60.90
Cloth	20.59%	47,563	24%	0.045	144.13
Rubber	0.00%	0	0%	0.045	0.00
Metal	0.00%	0	0%	0	0.00
Glass	0.70%	1,617	0%	0	4.90
Inert	1.71%	3,950	0%	0	11.97
Total	100.00%	231,000			700

Hence baseline emissions from Solid waste disposal site over the 1st crediting period would be as below which has been calculated in the submitted ER sheet and confirmed.

Year	Emission from Food (tCO ₂ e/year)	Emission from paper (tCO ₂ e/year)	Emission from wood including coconut shell (tCO ₂ e/year)	Emission from green mix (tCO ₂ e/year)	Emission from cloth (tCO ₂ e/year)	BE _{CH₄,SWDS,y} (tCO ₂ e/year)
1	0	5,721.8	385.2	9630.3	3586.4	19,324
2	0	11,191.9	761.0	18,654.5	7014.9	37,622
3	0	16,421.2	1127.4	27,110.9	10,292.6	54,952
4	0	21,420.5	1484.8	35,035.0	13,426.1	71,366
5	0	26,199.7	1833.4	42,460.5	16,421.7	86,915
6	0	30,768.7	2173.3	49,418.6	19,285.5	101,646
7	0	35,136.6	2504.9	55,938.9	22,023.2	115,604

Total Baseline Emissions from RDF combustion	487,430
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Baseline Emissions from Electricity Generation:

As per para 22 of AMS- I.D, version 18.0, the baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

PP has calculated the baseline emissions using equation (1) of AMS-I. D, version 18.0 as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where,

BE_y = Baseline emissions in year y (t CO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

The VCS PD was checked and it is confirmed that PP has corrected applied the relevant paragraphs and equations/parameters are correctly mentioned and hence acceptable.

To calculate Combined margin emissions factor i. e. $EF_{grid,CM,y}$, PP has used the latest version of tool 07 and same is found correct and assessment is presented below.

As per para 14 of Tool 7, version 07.0 the following steps have been followed by PP.

Step 1: Identify the relevant electricity systems;

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);

Step 3: Select a method to determine the operating margin (OM);

Step 4: Calculate the operating margin emission factor according to the selected method;

Step 5: Calculate the build margin (BM) emission factor;

Step 6: Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity systems

As described in tool “For determining the electricity emission factors, identify the relevant project electricity system. Similarly, identify any connected electricity systems”. It also states that “If the DNA of the host country has published a delineation of the project electricity system and connected electricity systems, these delineations should be used”. Keeping this into consideration, the Central Electricity Authority (CEA), Government of India has divided the Indian Power Sector into five regional grids viz. Northern, Eastern, Western, North-eastern and Southern.

However, since August 2006, all regional grids except the Southern Grid had been integrated and were operating in synchronous mode, i.e., at same frequency. Consequently, the Northern, Eastern, Western and North-Eastern grids were treated as a single grid named as NEWNE grid from FY 2007-08 onwards for the purpose of this CO2 Baseline Database. As of 31-December-2013, the Southern grid has also been synchronized with the NEWNE grid; hence forming one unified Indian Grid. Since the project supplies electricity to the Indian grid, emissions generated due to the electricity generated by the Indian grid as per CM calculations serves as the baseline for this project.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

As per the tool, project owners may choose between the following two options to calculate the operating margin and build margin emission factor:

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

The project owner has chosen only grid power plants in the calculation which is found acceptable.

Step 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step 4:

- Simple OM; or
- Simple adjusted OM; or
- Dispatch data analysis OM; or
- Average OM.

The data required to calculate Simple adjusted OM and Dispatch data analysis OM is not possible due to lack of availability of data to project Proponents. The choice of other two options for calculating operating margin emission factor depends on generation of electricity from low-cost/must-run sources. In the context of the methodology low cost/must run resources typically include hydro, geothermal, Solar, low-cost biomass, nuclear and solar generation.

Share of Must Run (Hydro/Nuclear) (% of Net generation)					
India	2018-19	2019-20	2020-21	2021-22	2022-23
	14.5%	17.0%	16.5%	15.81%	15.33%

Source: Central Electricity Authority (CEA) Database Version 19, 2023³²

The above data clearly shows that the percentage of total grid generation by low-cost/ must-run plants (on the basis of average of five most recent years) for the Indian grid is less than 50 % of the total generation. Thus, the Average OM method cannot be applied, as low cost/must run resources constitute less than 50% of total grid generation.

³² <https://cea.nic.in/cdm-co2-baseline-database/?lang=en>

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost/must-run power plants/units.

For the simple OM, the simple adjusted OM and the average OM, the emissions factor can be calculated using either of the two following data vintages:

- (a) Ex-ante option: if the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

OR,

- (b) Ex-post option: if the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring.

PP has chosen ex-ante option for calculation of Simple OM emission factor using a 3-year generation-weighted average, based on the most recent data available at the time of submission of the PD to the VCS verifier for verification. OM determined at validation stage will be the same throughout the crediting period. There will be no requirement to monitor & recalculate the emission factor during the crediting period.

Step 4: Calculate the operating margin emission factor ($EF_{grid,OM\ Simple,y}$) according to the selected method:

The operating margin emission factor has been calculated using a 3-year data vintage:

Net Generation in Operating Margin (GWh) (incl. Imports)			
INDIAN Grid	2020-2021	2021-2022	2022-2023
	958,218	1,035,672	1,117,846
Simple Operating Margin (tCO ₂ /MWh) (incl. Imports)			
INDIAN Grid	2020-2021	2021-2022	2022-2023
	0.9402	0.9605	0.9710

Weighted Generation Operating Margin	
INDIAN Grid	0.9580

STEP 5: Calculate the build margin emission factor ($EF_{BM,y}$):

Option 1 as described above is chosen to calculate the build margin emission factor for the project activity. BM is calculated ex-ante based on the most recent information available from CEA Database version 19.0 at the time of submission of PDD and is fixed for the entire crediting period.

Build Margin (tCO ₂ /MWh) (not adjusted for imports)	
INDIAN Grid	2022-2023
	0.8670

STEP 6: Calculate the combined margin (CM) emissions factor:

Combined Margin – The combined margin is the weighted average of the simple operating Margin and the build margin. In particular, for intermittent and non-dispatch able generation types such as wind and solar photovoltaic, Tool 07- tool to calculate the emission factor for an electricity system, Version 07.0.0, allows to weigh the operating margin and Build margin at 75% and 25%, respectively for wind and solar projects and 50% and 50%, respectively for hydro and biomass projects.

As per para 85, equation (16) of Tool 07, version 07.0, the combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Where,

$EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (t CO₂/MWh)

$EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (t CO₂/MWh)

w_{OM} = Weighting of operating margin emissions factor (percent)

w_{BM} = Weighting of build margin emissions factor (percent)

Hence, Combined Margin Emission Factor ($EF_{grid,CM,y}$) = $0.50 \times 0.9580 + 0.50 \times 0.8670 = 0.9125$ tCO₂/MWh

VVB has checked and confirmed that the calculation of emission factor is appropriately described and found correct.

The expected power generation from the project is calculated as below:

Total power generation	= $14 \times 80\% \times 330 \times 24$
	= 88,704 MWh/year
Auxiliary Power Consumption	= 12.5%
Net Power Generation	= $88,704 \times (1 - 0.125)$
	= 77,616 MWh/year

With this expected power generation and using the above emission factor, the total baseline emissions would be;

Total Baseline Emissions	= $77,616 \times 0.9125$
	= 70,824 tCO ₂ e/year

The cumulative baseline emissions would be;

Baseline emissions from electricity generation = 70,824 tCO₂e/year

+

Baseline emissions from methane avoidance = 69,632 tCO₂e/year

Hence, total baseline emissions from waste disposal and electricity generation = 122,899 tCO₂e/year

VVB has checked the calculations presented in the ER sheet and same were found appropriate.

Project emissions:

As per para 22 of AMS- III. E, " Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment ", version 17.0, the project emissions consists of

$$PE_y = PE_{y,comb} + PE_{y,transp} + PE_{y,power}$$

Where,

PE_y = Project activity direct emissions in the year y (t CO₂e)

$PE_{y,comb}$ = Emissions through combustion and gasification of non-biomass carbon of waste and RDF/SB in the year y (t CO₂e)

$PE_{y,transp}$ = Emissions through incremental transportation in the year y (t CO₂e)

$PE_{y,power}$ = Emissions through electricity or diesel consumption in the year y (t CO₂e)

Project emissions from combustion process of RDF:

As per para 24 of AMS- III.E, "Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment", version 17.0, the project emissions from combustion of the non-biomass (i.e., fossil) carbon content of the wastes and RDF/SB and from the auxiliary fossil fuel consumed will be estimated as follows:

$$PE_{y,comb} = Q_{y,non-biomass} \times 44/12 + Q_{y,fuel} + EF_{y,fuel}$$

Where,

$Q_{y,non-biomass}$ = Non-biomass carbon of the waste and RDF/SB combusted/gasified in the year y (tonnes of carbon)

$Q_{y,fuel}$ = Quantity of auxiliary fossil fuel used in the year y (tonnes)

$EF_{y,fuel}$ = CO₂ emission factor for the combustion of the auxiliary fossil fuel (tonnes CO₂ per tonne fuel, according to latest IPCC Guidelines)

The estimation of project emission from combustion of RDF is as follows:

$$Q_{non-biomass} = W_{j,x} \times \text{Default dry matter content (\%)} \times FCC_{j,y} \times FFC_{j,y}$$

Where,

$W_{j,x}$ = Quantity of waste type j fed into combustor c the in year x (dry basis)
 $FCC_{j,y}$ = Fraction of total carbon content in waste type j in year x
 $FFC_{j,y}$ = Fraction of fossil carbon in total carbon content of waste type j in year x (weight fraction)

Description	W_{jx} (MT)	Default wet matter content	Default dry matter content	W_{jx}	$FCC_{j,y}$	$FFC_{j,y}$	$Q_{y, non,biomass}$ (MT)
j	TPY (Wet)	(%)	(%)	TPY (Dry)	Total Carbon (%)	Fossil carbon in total carbon (%)	Non-biomass carbon of the RDF (TPY)
Food	0	60%	40%	0	38%	0%	0
Paper	45,530	10%	90%	40,977	46%	1%	188
Wood including Coconut Shell	5,082	15%	85%	4,320	50%	0%	0
Green/Mix	107,161	60%	40%	42,864	49%	0%	0
Plastic	20,097	0%	100%	20,097	75%	100%	15,073
Cloth	47,563	20%	80%	38,050	50%	20%	3,805
Rubber	0	16%	84%	0	67%	20%	0
Metal	0	0%	100%	0	0%	0%	0
Glass	1,617	0%	100%	1,617	0%	0%	0
Inert	3,950	10%	90%	3,555	3%	100%	107
Total	231,000			151,481			19,173

Parameter	Description	Unit	Value
$Q_{non,biomass}$	Non-biomass carbon of the waste and RDF/SB combusted/gasified in the year y (tonnes of carbon)	tonnes	19,173
Q_{fuel}	Quantity of auxiliary fossil fuel used in the year y	litres	29,232
$\rho_{i,y}$	Density of Diesel	kg/litre	0.88

$Q_{y,fuel}$	Quantity of auxiliary fossil fuel used in the year y	tonnes	25.724
$EF_{y,fuel}$	Weighted average CO ₂ emission factor of fuel (diesel)	Kg CO ₂ /TJ	74,800
$NCV_{i,y}$	Net calorific value of fuel (diesel)	TJ/Gg	43.3
$EF_{y,fuel}$	CO ₂ emission factor for the combustion of the auxiliary fossil fuel (tonnes CO ₂ per tonne fuel, according to latest IPCC Guidelines)		3.24

$$PE_{y,comb} = Q_{y,non-biomass} \times \frac{44}{12} + Q_{y,fuel} \times EF_{y,fuel}$$

$$\text{Total project emissions from combustion of RDF (PE}_{y,comb}) = (19,173 \times 44/12) + (25.724 \times 3.24) \\ = 70,384 \text{ tCO}_2\text{e/year}$$

Project Emissions from fossil fuel combustion

As per para 6, equation (1) of tool 03, version 03.0, the project emissions through diesel consumption are determined as follows:

$$PE_{FC,j,y} = \sum_i FC_{i,j,y} \times COEF_{i,y}$$

Where,

$PE_{FC,j,y}$	= CO ₂ emissions from fossil fuel combustion in process j during the year y (tCO ₂ /yr)
$FC_{i,j,y}$	= Quantity of fuel type i combusted in process j during the year y (mass or volume unit/yr)
$COEF_{i,y}$	= CO ₂ emission coefficient of fuel type i in year y (tCO ₂ /mass or volume unit)

As per para 7 (b), equation (4) of tool 03, version 03.0, the CO₂ emission coefficient $COEF_{i,y}$ is calculated based on net calorific value and CO₂ emission factor of the fuel type i, as follows:

$$COEF_{i,y} = NCV_{i,y} \times EF_{CO2,i,y}$$

Where,

$NCV_{i,y}$	= Weighted average net calorific value of the fuel type i in year y (GJ/mass or volume unit)
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$FC_{i,j,y}$	Quantity of fossil fuel combusted in the project activity (diesel)	Litres	4080
$\rho_{i,y}$	Density of Diesel	kg/litre	0.88
$FC_{i,j,y}$	Quantity of fossil fuel combusted in the project activity	tonnes	3.6
$NCV_{i,y}$	Net calorific value of fuel (diesel)	TJ / Gg	43.3

$EF_{CO_2,i,y}$	Weighted average CO ₂ emission factor of fuel (diesel)	Kg/TJ	74800
$COEF_{i,y}$	CO ₂ emission factor for the combustion of the fossil fuel		3.24

$EF_{CO_2,i,y}$ = Weighted average CO₂ emission factor of fuel type i in year y (tCO₂/GJ)

The estimation of project emission from fossil-fuel combustion is as follows:

Project Emissions from fossil fuel combustion = (3.6 *3.24)
= 11 tCO₂e/year

Project Emissions from Incremental Transportation

$$PE_{TR,m} = \sum_f D_{f,m} \times FR_{f,m} \times EF_{CO_2,f} \times 10^{-6}$$

Where,

$PE_{TR,m}$ = Project emissions from transportation of freight monitoring period m (t CO₂)

$D_{f,m}$ = Return trip distance between the origin and destination of freight transportation activity f in monitoring period m (km)

$FR_{f,m}$ = Total mass of freight transported in freight transportation activity f in monitoring period m (t)

$EF_{CO_2,f}$ = Default CO₂ emission factor for freight transportation activity f (g CO₂/t km)

f = Freight transportation activities conducted in the project activity in monitoring period m

Parameter	Description	Unit	Value	Remark
$D_{MSW,m}$	Return trip distance between the origin and destination of freight transportation for collection of MSW	km	0	The collection and transportation of MSW is under the scope of Pimpri- Chinchwad Municipal Corporation. The average distance between the project site and waste collection point is in same range of distance between the waste collection point and the landfill site in the baseline scenario. Hence the project emissions from freight transportation for collection of MSW is considered as zero.
$FR_{MSW,m}$	Total mass of freight transported in freight transportation for collection of MSW	t	700	
N	No of Operating days	days	330	
$D_{Inert,m}$	Return trip distance between the origin and destination of freight	km	0	The inert waste has been disposed in the Landfill which is next to the project site and hence,

	transportation for disposal of inert waste			the return trip distance between the origin and destination of freight transportation for disposal of inert waste is considered as zero
FR_{inert,m}	Total mass of freight transported in freight transportation for disposal of inert waste	TPD	142.85	Plant Record
D_{ash,m}	Return trip distance between the origin and destination of freight transportation for disposal of combustion residue	km	0	The fly ash and bottom ash has been sent to Construction & Demolition waste processing plant. Hence, the return trip distance between the origin and destination of freight transportation for disposal of combustion residue is zero.
F_{ash,m}	Total mass of freight transported in freight transportation for disposal of combustion residue	TPD	77.38	Plant Record
EF_{CO2,f}	Default CO ₂ emission factor for freight transportation activity f	g CO ₂ /t km	245	As per table 1 of tool 12, version 01.1.0
PE_{y,transp}	Project Emissions from Incremental Transportation	tCO₂/year	0	Calculated

Leakage emissions:

According to the applied methodology of AMS- III.E, “Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment” *version 17.0*, para 37, the RDF produced is not subjected to anaerobic conditions before its combustion. Hence, no leakage emission is considered for incineration of RDF.

Net emission reductions:

As per paragraph 38 of AMS III.E, *version 17.0*, The emission reduction achieved by the project activity will be measured as the difference between the baseline emission and the sum of the project emission and leakage.

$$ER_y = BE_y - (PE_y - LE_y) \quad \text{-- equation (7)}$$

Where,

ER_y = Emission reductions in year y (t CO₂)

BE_y = Baseline Emissions in year y (t CO₂)

PE_y = Project emissions in year y (t CO₂)

LE_y = Leakage emissions in year y (t CO₂)

As per paragraph 43 of AMS I.D *version 18.0*, the emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y \quad \text{-- equation (9)}$$

Where:

ER_y = Emission reductions in year y (t CO₂)

BE_y = Baseline Emissions in year y (t CO₂)

PE_y = Project emissions in year y (t CO₂)

LE_y = Leakage emissions in year y (t CO₂)

VVB confirms that the methodology and the referenced tools are applied correctly to calculate baseline emissions, project emissions, leakage and net GHG emission reductions and removals.

PP has provided excel datasheet which provides complete calculation for baseline, project and leakage emissions. Same was checked to confirm that the values considered are correct and equations are used in line with the applied methodology and tools. Based on the submitted excel datasheet, VVB checked and confirmed that the values presented for emissions is correct in the below table for the complete 1st crediting period.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
07-October-2023 to 31-December-2023	20,961	16,586	0	4,374
01-January-2024 to 31-December-2024	93,081	70,395	0	22,685
01-January-2025 to 31-December-2025	110,076	70,395	0	39,681
01-January-2026 to 31-December-2026	126,184	70,395	0	55,788
01-January-2027 to 31-December-2027	141,388	70,395	0	70,992
01-January-2028 to 31-December-2028	155,946	70,395	0	85,550
01-January-2029 to 31-December-2029	169,586	70,395	0	99,191

01-January-2030 to 06-October-2030	137,304	53,809	0	83,495
Total in (7 years)	954,525	492,765	0	461,756
Annual Average	119,315	61,595	0	57,719

3.3.7 Methodology Deviations

For the current validation, PP has not identified any methodology deviations applied to the project.

3.3.8 Monitoring Plan

To calculate the emission reductions, the following parameters are fixed ex-ante as per the approved CDM baseline and monitoring methodology AMS III. E, version 17.0 and AMS. I.D, version 18.0 and other tools used.

Parameters determined ex-ante: -

- GWP_{CH4} – The ex-ante value of 28 tCO₂e/tCH₄ for the parameter ‘Global Warming Potential for methane’ is based on 2019 refinements to 2006 IPCC Guidelines for National Greenhouse Gas Inventories, 5th assessment report which is the most recent one and found acceptable.
- EF_{grid,OM,y} – The ex-ante value of 0.9580 tCO₂e/MWh for the parameter ‘Operating Margin CO₂ emission factor in year y’ is based on CO₂ emission database, version 19.0, November, 2023 published by Central Electricity Authority (CEA), Government of India. The value is based on most recent 3 year data from CEA database i. e. (2020-2021, 2021-2022, 2022-2023) and found acceptable.
- EF_{grid,BM,y} – The ex-ante value of 0.8670 tCO₂e/MWh for the parameter ‘Build Margin CO₂ emission factor in year y’ is based on CO₂ emission database, version 19.0, November, 2023 published by Central Electricity Authority (CEA), Government of India. The value is based on most recent year data (2022-2023) from CEA database and found acceptable.
- EF_{grid,CM,y} – The ex-ante value of 0.9125 tCO₂e/MWh for the parameter ‘Combined Margin CO₂ emission factor in year y’ is based on CO₂ emission database, version 19.0, November, 2023 published by Central Electricity Authority (CEA), Government of India. The value calculated based on operating and build margin value calculated from CEA database and found acceptable.
- Ø_{default} – The ex-ante value of 0.80 for the parameter ‘Default value for the model correction factor to account for model uncertainties’ is for tropical humid/wet climate conditions as per the Tool “Emissions from solid waste disposal site” for Application B. VVB has confirmed humidity and wet climate condition at the project location on the basis of the data available from the local database such as [Indian Climate - Climate of Pune](#) which confirms that average temperature is above 20°C and https://www.pcmcindia.gov.in/residents_info.php which confirms that average

annual rainfall is less than 1000 mm thus satisfying the requirements of table 1 of tool 04, version 08.1 and hence found acceptable.

- f. F – The ex-ante value of 0.5 for the parameter ‘Fraction of methane in the SWDS gas (volume fraction)’ is as per IPCC 2006 Guidelines for National Greenhouse Gas Inventories and is a default value and found acceptable.
- g. OX – The ex-ante value of 0.1 for the parameter ‘Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)’ is as per the Tool “Emissions from solid waste disposal site” and found acceptable.
- h. $DOC_{f,default}$ – The ex-ante value of 0.5 for the parameter ‘Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS’ is as per IPCC 2006 Guidelines for National Greenhouse Gas Inventories and found acceptable.
- i. $MCF_{default}$ – The ex-ante value of 1.0 for the parameter ‘methane correction factor’ is used as per IPCC 2006 Guidelines for National Greenhouse Gas Inventories for anaerobic managed solid waste disposal sites. VVB noted that the SWDS is above the water table and is managed solid waste disposal sites, includes controlled placement of waste and mechanical compacting and levelling, as per the Tool “Emissions from solid waste disposal sites”. Hence found acceptable.
- j. DOC_j – The ex-ante value of the parameter ‘Fraction of degradable organic carbon in the waste type j (weight fraction)’ are adapted from Volume 5, Tables 2.4 and 2.5 of IPCC 2006 Guidelines for National Greenhouse Gas Inventories as per the Tool “Emissions from solid waste disposal sites”. Hence found acceptable.
- k. K_j – The ex-ante value of the parameter ‘Decay rate for the waste type j’ for different waste types are adapted from Volume 5, Tables 3.3 of IPCC 2006 Guidelines for National Greenhouse Gas Inventories as per the Tool “Emissions from solid waste disposal sites”. Hence found acceptable.
- l. FCC_j – The ex-ante value of the parameter ‘Fraction of total carbon in total carbon content of waste type j’ for different waste types are adapted from Volume 5, chapter 2, Table 2.4 of IPCC 2006 Guidelines for National Greenhouse Gas Inventories as per the Tool “Emissions from solid waste disposal sites”. Hence found acceptable.
- m. Dry matter content in % of wet weight - The ex-ante values of the parameter are adapted from Volume 5, chapter 2, Table 2.4 of IPCC 2006 Guidelines for National Greenhouse Gas Inventories as per the Tool “Emissions from solid waste disposal sites”. Hence found acceptable.
- n. $NCV_{i,y}$ – The ex-ante value of 43.3 TJ/Gg for the parameter ‘Weighted average net calorific value of fuel type i in year y’ for the fuel – diesel is sourced from Table 1.2 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories and found acceptable.
- o. $EF_{CO2,i,y}$ – The ex-ante value of 74,800 Kg/TJ for the parameter ‘Weighted average CO₂ emission factor of fuel type i in year y’ for the fuel – diesel is sourced from Table 1.4 of Chapter 1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories and found acceptable.

- p. $\rho_{i,y}$ - The ex-ante value of 880 kg/m³ for the parameter 'Weighted average density of fuel type i in year y' is based on data published by Indian Oil Corporation Limited (IOCL) as a local value usage is permitted and same is checked and found acceptable.
- q. $EF_{CO_2,f}$ - The ex-ante values for the parameter are based on default CO₂ emission factors take into account emissions generated by loaded outbound trips and empty return trips as per the applied tool and hence found acceptable.

Parameters determined ex-post: -

- a) $EG_{Pj,y}$ - Quantity of net electricity generation supplied by the project (Waste to Energy) plant/unit to the grid in year y: This parameter is monitored continuously and calculated based on export by project and import energy by the project to arrive at the net electricity supply to the grid in MWh/year based on joint meter readings (JMR) by PP and DISCOM officials. PP has installed bi-directional energy meters with accuracy class of 0.2s. There are 02 meters – Main and Check so in event of failure of any one meter, the other can be considered to get the values of energy exported/imported. Energy meters Q0833070 and Q0833075 ("Standard Summator-100") of accuracy class 0.2s.

VVB noted that both the energy meters are calibrated as on date based on the calibration certificates. As per local government guidelines in India, calibration of energy meter is to be conducted at least once in the 5 years³³. The generated energy based on JMR can be cross checked against invoices raised by PP to the DISCOM in line with agreement signed with them.

- b) Electricity Consumed by the project activity - Quantity of net electricity consumed by the project activity from the grid in year y : This parameter is measured based on energy imported by the project recorded at energy meter installed at Substation in MWh/year and monitored continuously. PP has installed bi-directional energy meters with accuracy class of 0.2s. There are 02 meters – Main and Check so in event of failure of any one meter, the other can be considered to get the values of energy imported. Energy meters Q0833070 and Q0833075 ("Standard Summator-100") of accuracy class 0.2s.

VVB noted that both the energy meters are calibrated as on date based on the calibration certificates.

- c) W_x - Total amount of waste disposed in a SWDS in year x: Monitoring frequency: Continuously, aggregated at least annually for year x. The measured value is multiplied by dry matter as per the methodology requirements. The weighbridge used for weighing is regularly calibrated in accordance with the manufacturer's specification i.e. annual as confirmed during the on-site visit. As confirmed during site visit and calibration records, calibration of electronic weighbridge will be done annually. Plant record for total solid waste reached at site maintained at site electronically as well as manually and are measured on daily basis.

³³ https://cea.nic.in/wp-content/uploads/2020/02/meter_reg.pdf

- d) $P_{n,j,X}$ - Weight fraction of the waste type j in the sample n collected during the year x . This parameter is monitored to know the waste composition in line with applied methodology and measured on sampled basis quarterly. As discussed during the on-site visit, random samples are collected after the pre-processing section i.e. from the MRF, out of which representative sample of more than 25 kg will be taken for analysis. The waste composition based on the waste types j are segregated and weighed for each waste fraction (measured on wet basis). The sampling records for 03 samples conducted on quarterly basis will be maintained by PP. same was checked and found consistent during the on-site inspection by the VVB.
- e) Z_x - Number of samples collected during the year x : In line with requirement of the methodology, PP will sample minimum 03 nos. from each of the - truck load, pre-processing area, boiler feed, star-screen rejects once in three months. This is found acceptable.
- f) $Q_{y,y}$ - Amount of waste combusted, gasified or mechanically/thermally treated by the project activity in each year y : The parameter is measured continuously in tonnes per day (TPD) which is the quantity of RDF fed to the boiler and recorded on daily basis and shall be aggregated monthly which is found acceptable. The quantity of RDF consumed shall be monitored on daily basis. Moreover, the load cells in the grab crane have been subjected to periodic calibration (in accordance with the manufacturers specification). Same was checked on site and was found consistent by VVB.
- g) $Q_{y,ash}$ - Quantity of combustion residues generated by the project activity in each year y : As a part of combustion of processed MSW i. e. RDF, the residual waste is generated which contains bottom ash and fly ash. As observed during on-site visit, the ash generated is weighed at the weighbridge and then dispatched to brick manufacturers on daily basis and recorded in plant records. The weight bridge used is calibrated and is in continual operation.
- h) $Q_{y,non-biomass}$ - Quantity of non-biomass carbon content of the RDF combusted in each year y : This parameter is monitored to understand non-biomass content of the waste combusted in line with applied methodology and measured on sampled basis quarterly. As discussed during the on-site visit, random samples will be collected after the pre-processing section i.e. from the vibro feeder, out of which representative sample of 25 kg will be taken for analysis. The sampling records for 03 samples conducted on quarterly basis will be maintained by PP. The quantity of non-biomass carbon content of the RDF will be calculated based on the IPCC default values of total carbon content (FCC) and fossil carbon content (FFC).
- i) CT_y - Average truck capacity for transportation: This parameter values in tonnes/truck are monitored via the recordings of the net capacity of the trucks at weighbridge readings and cross-checked with the vehicle licenses. Plant records are maintained for each truck at site electronically as well as manually and are measured on daily basis.
- j) $Q_{y,fuel}$ - Quantity of auxiliary fossil fuel used in the year y : As discussed during on-site visit, PP uses Diesel as a start-up fuel for boiler. Same is measured in tonnes/year. The plant records of diesel consumption will be cross checked against purchase bills.

- k) $FC_{i,j,y}$ - Quantity of fuel type i combusted in process j during the year y : As noted during the on-site visit, PP uses vehicles within the project boundary for waste transfer from trucks to bunker from bunker to vibro feeder. PP has also installed 01 Diesel Generation set. Amount of fuel (Diesel) used will be measured in tonnes/year. The plant records of diesel consumption will be cross checked against purchase bills.
- l) $D_{inert,m}$ - Return trip distance between the origin and destination of freight transportation activity for disposal of inert waste in monitoring period m : PP will measure and record average distance travelled by the truck through the odometer readings of trucks. The plant records will have the data in km on daily basis for this parameter.
- m) $D_{f,m}$ - Return trip distance between the origin and destination of freight transportation activity for disposal of ash in monitoring period m : PP will measure and record average distance travelled by the truck through the odometer readings of trucks. The plant records will have the data in km on daily basis for this parameter.
- n) $FR_{inert,m}$ - Total mass of freight transported in freight transportation activity for disposal of inert waste in monitoring period m : The parameter will be monitored in daily basis via the recordings of the net capacity of the trucks at weighbridge readings and cross-checked with the vehicle licenses.
- o) $F_{ash,m}$ - Total mass of freight transported in freight transportation activity for disposal of ash in monitoring period m : The parameter will be monitored in daily basis via the recordings of the net capacity of the trucks at weighbridge readings and cross-checked with the vehicle licenses.

The monitoring plan is designed to ensure accurate data collection, recording, and maintenance according to the AMS III.E and AMS I.D methodologies. A dedicated team, led by a senior manager, is responsible for overseeing the project, with specific roles for data collection, supervision, and internal auditing. Data on various parameters like MSW composition, electricity, fuel consumption, and combustion residues are continuously monitored and recorded. Training is provided to personnel on operational procedures, safety protocols, and data management. The plan includes emergency preparedness measures and a systematic internal auditing process to ensure compliance, efficiency, and data accuracy. Quality Assurance and Quality Control (QA/QC) procedures, including regular calibration of equipment, are implemented to maintain the reliability of the monitoring system.

VVB concludes that the monitoring plan adheres to the requirements of the applied methodology and referenced tool and found correct.

3.4 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VALIDATION OPINION

4.1 Validation Summary

Applus+ Certification has been contracted by Antony Lara Renewable Energy Pvt. Ltd. (ALREPL) to validate the "14 MW Waste-to-Energy Project by Antony Lara Renewable Energy Pvt. Ltd" undertaken by ALREPL. The project aims to generate renewable energy while reducing greenhouse gas (GHG) emissions through sustainable waste management practices. The project proponent is responsible for compiling and reporting the GHG emissions data and estimating the GHG emissions reductions according to the project's Monitoring Plan. This plan follows the Approved Consolidated Methodologies AMS.III-E Version 17.0, which pertains to the avoidance of methane production through controlled combustion, gasification, or mechanical/thermal treatment, and AMS.I-D Version 18.0, related to grid-connected renewable electricity generation.

The management of Antony Lara Renewable Energy Pvt. Ltd. (ALREPL) is responsible for the preparation of the GHG emissions data and the reported expected GHG emissions reductions on the basis set out within the monitoring plan in the PD. Thus, GHG statement is responsibility of Antony Lara Renewable Energy Pvt. Ltd. (ALREPL) who is the project proponent.

The validation process conducted by Applus+ Certification adhered to the Kyoto Protocol guidelines and the Voluntary Carbon Standard (VCS) board's requirements. The approach was risk-based, focusing on understanding the risks associated with GHG emissions data estimation and the controls implemented to mitigate these risks. During the validation, the team raised 03 Corrective Action Requests (CARs) and 05 Clarification Requests (CLs), all of which were successfully addressed by the project proponent.

The validation team has confirmed that the project complies with the VCS Standard Version 4.7 and meets all relevant VCS requirements. The methodologies AMS.III-E Version 17.0 and AMS.I-D Version 18.0 have been correctly applied to the project. The project's baseline has been thoroughly justified, and the

calculation of emission reductions is conducted transparently and conservatively. The estimated average annual emission reductions amount to 57,719 tCO₂e, with a total of 461,756 tCO₂e expected over the 7-year (twice renewable) crediting period.

No restrictions or uncertainties were identified during the validation process. Applus+ Certification has conducted the validation of the GHG statement in accordance with ISO 14064-3:2019, ensuring that the project adheres to international standards for GHG reporting and validation. Consequently, Applus+ Certification requests the registration of the project as a VCS project activity, recognizing its potential to contribute significantly to GHG emission reductions and sustainable energy generation.

4.2 Validation Conclusion

Our validation approach adhered to the applicable regulatory requirements, including the VCS Standard version 4.7, relevant VCS regulatory documents and templates (as detailed in Appendix 2 of this Validation Report), and, where applicable, the rules and requirements of the UNFCCC, as well as associated decisions by the VCS Secretariat and the UNFCCC Executive Board. This approach was risk-based, focusing on understanding the risks linked to the estimated GHG emissions data and the measures implemented to mitigate these risks.

The validation confirms that the VVB team has verified the data and information in the GHG statement, ensuring they are accurately projected. All assumptions have been substantiated by available documentation, third-party reports, or references listed in Appendix 2. While the estimates are grounded in current assumptions, they remain subject to change. The VVB concludes that the project is likely to achieve the estimated emission reductions as declared by the Project Proponent. Additionally, the project's baseline and additionality have been assessed against the methodologies "AMS.III-E Version 17.0" and "AMS.I-D Version 18.0," with the project's monitoring plan similarly evaluated against these methodologies. A risk-based approach was consistently applied throughout the validation process. The review of the project description, baseline, and monitoring methodology documents, along with background investigations and follow-up interviews with the Project Proponent, provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to issue a positive validation opinion in accordance with VCS requirements. Ultimately, based on the assessment of the data, assumptions, limitations, and methods, it is concluded that the proposed is likely to achieve the estimated GHG emission reductions.

Thus, VVB concludes that the project is likely to achieve estimated emission reductions as described below:

Crediting period: Starts on 07-October-2023 and ends on 06-October-2030.

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
07 - October-2023 to 31-December-2023	20,961	16,586	0	4,374	0	4,374
01 - January-2024 to 31-December-2024	93,081	70,395	0	22,685	0	22,685
01 - January-2025 to 31-December-2025	110,076	70,395	0	39,681	0	39,681
01 - January-2026 to 31-December-2026	126,184	70,395	0	55,788	0	55,788
01 - January-2027 to 31-December-2027	141,388	70,395	0	70,992	0	70,992
01 - January-2028 to 31-December-2028	155,946	70,395	0	85,550	0	85,550
01 - January-2029 to 31-December-2029	169,586	70,395	0	99,191	0	99,191

01 - January-2030 to 06-October-2030	137,304	53,809	0	83,495	0	83,495
Total in (7 years)	954,525	492,765	0	461,756	0	461,756
Annual Average	119,315	61,595	0	57,719	0	57,719

The non-permanence risk rating (%)	NA
If applicable, the Long-term Average (LTA), whether it has been properly updated, and if it has been reached.	NA

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

Not applicable

APPENDIX 2: DOCUMENT REFERENCES

Sr. No	Author	Title	References to the document	Provider
1	Project Proponent	Initial VCS PD Final VCS PD	Version – 01 dated 22-January-2024 Version – 08 dated 22-April-2025	Project Proponent
2	Project Proponent	Initial ER Sheet Final ER sheet	Version – 02 dated 22-January-2024 Version – 06 dated 17-March-2025	Project Proponent
3	Project Proponent	Initial IRR Sheet Final IRR sheet	Version – 02 dated 22-January-2024 Version – 04 dated 22-April-2024	Project Proponent
4	UNFCCC	AMS III.E.: Avoidance of methane production from decay of biomass through controlled combustion, gasification or mechanical/thermal treatment AMS I.D: Grid connected renewable electricity generation	Version 17.0 Version 18.0	

5	UNFCCC	CDM Tools: Tool 03- Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion Tool 04- Tool to calculate emissions from solid waste disposal sites Tool 07- Tool to calculate the emission factor for an electricity system Tool 12- Project and leakage emissions from transportation of freight Tool 21- Demonstration of additionality of small-scale project activities Tool 27- Investment analysis	Version 03.0 Version 08.1 Version 07.0 Version 01.1.0 Version 13.1 Version 08.0 and Version 14.0	
6	VERRA	VCS Requirements: Verified Carbon Standard Program Guide Verified Carbon Standard VCS Program Definitions VCS Registration and Issuance Process VCS Validation and Verification Manual VCS Project Description template VCS Validation Report Template	Version 4.4 Version 4.7 Version 4.4 Version 4.4 Version 3.2 Version 4.3 Version 4.3	VERRA
7	Project Proponent	Environmental and Social Impact Assessment (ESIA) for PCMC Waste to Energy Project	October, 2021	Project Proponent
8	Project Proponent	Piping and Instrumentation diagram Block diagram / Line diagram	10/01/2018	Project Proponent
9	Project Proponent	Waste characterization report for the waste received by the project	26/03/2022 24/02/2023	Project Proponent

10	Pimpri Chinchwad Municipal Corporation	Certificate of start of commercial operation (started generating power) on 07/10/2023	Dated 06-February-2024	Project Proponent
11	Maharashtra Pollution Control Board	Consent to Establish Consent to Operate	Dated 06-January-2020 Dated 11-July-2023	Project Proponent
12	Pimpri Chinchwad Municipal Corporation	Building permission approval for WTE plant	Dated 11-November--2019	Project Proponent
13	Government of Maharashtra (metrology dept.)	Weighbridge calibration report Grate crane calibration report	Dated 04-February-2023 and 31-March-2024 Dated 10-April-2023	Project Proponent
14	Maharashtra Electricity Distribution Company Limited	Commissioning of 14 MW Waste to Energy project for power evacuation at 22 kV level of 220 kV Bhosari sub station	Dated 06-October-2024	Project Proponent
15	Government of Maharashtra (Industrial Health and safety dept.)	Registration of factory and permission to operate the same for WTE plant	Dated 16-July-2023	Project Proponent
16	ESBEE Power Solutions Pvt. Ltd.	Detailed Project Report for 14 MW Waste to Energy project	December, 2017	Project Proponent
17	Project Proponent	Concession agreement between PCMC and ALREPL Conversion charges agreement between PCMC and ALREPL	Dated 06-October-2020 Dated 26-June-2020	Project Proponent
18	Project Proponent	Supply and service agreement between ALREPL and Hitachi Zosen Private Limited for 700 TPD waste to energy unit	Dated 06-July-2020	Project Proponent
19	Pimpri Chinchwad Municipal Corporation	Request for proposal for waste to energy project	Dated 25-September-2017 Dated 28-February-2018	Project Proponent

		Letter of Acceptance / Letter of award to ALREPL		
20	Project Proponent	Technical specifications of Boiler, Incinerator, Flue gas treatment unit, ID Fan, Steam turbine, Generator, air cooled condenser etc.		Project Proponent
21	Project Proponent	Electricity generation records for year 2023-24 along with the monthly invoices raised against the same Energy meter initial calibration certificate by Secure Meters Limited	October , 2023 to March, 2024 21-July-2023	Project Proponent
22	Power Finance Corporation Limited	Loan sanction letter	01-November-2019	Project Proponent
23	Project Proponent	Local stakeholder consultation report including, Invitation Letter, Public Notice, Attendance sheet, Feedback Form, Minute of meetings Contact information displayed on company display board including phone, number, E-mail of project manager for stakeholders and copy grievance register placed at the entry gate.	03-November-2023	Project Proponent
24	Project Proponent	1.Health and Safety register, 2.Plant log books for a.electricity generation b.waste- incoming waste and load cell values, both c.ash generation and disposal (fly ash and bottom ash) d.Auxiliary fossil fuel consumption e.Plant breakdown/shutdown records 3. Company HR policy 4. Health and Safety policy 5. Anti-discrimination and anti-harassment policy		Project Proponent

25	Project Proponent	Employment records for the months - October, 2023 to February, 2024 Salary slips on sample basis for the months - October, 2023 to February, 2024 Employee training records for the months - October, 2023 to February, 2024		Project Proponent
26	CDM	<u>CDM Website</u> https://cdm.unfccc.int/Projects/projectsearch.html last assessed on 09-November-2023	-	Publicly available
27	VERRA	<u>Verra Registry</u> https://registry.verra.org/app/search/VCS/All%20Projects last assessed on 09-November-2023	-	Publicly available
28	Gold Standard	<u>GS4GG Website:</u> https://registry.goldstandard.org/projects?q=&page=1 last assessed on 09-November-2023		Publicly available
29	I-REC Standard	<u>International REC Standard (I-REC)</u> https://www.irecstandard.org/registries/ last assessed on 09-November-2023	-	Publicly available
30	Project Proponent	No double counting declaration	Dated 28-May-2024	Project Proponent
31	Project Proponent	GST rates: <u>Goods & Service Tax, CBIC, Government of India :: GST Goods and Services Rates (cbic-gst.gov.in)</u> Tariff policy - <u>05.pdf (cercind.gov.in)</u> Income tax rate: <u>Depreciation rates (incometaxindia.gov.in)</u> / <u>Income Tax Department (incometaxindia.gov.in)</u>		Project Proponent
32	Project Proponent	MSW / RDF Records maintained at the site with respect to its usage Fly ash/ bottom ash records	-	Project Proponent
33	Project Proponent	Physical / chemical analysis conducted on sampling basis for MSW received and RDF produced		Project Proponent

34	Government of India	Central Electricity Authority (CEA) database version 19.0	December, 2023	Government of India
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APPENDIX 3: ABBREVIATIONS

Abbreviations	Full text
ALREPL	Antony Lara Renewable Energy Pvt. Ltd.
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CEA	Central Electricity Authority
CERC	Central Electricity Regularity Commission
CL	Clarification request
CO ₂ e	Carbon dioxide equivalent
CTE	Consent to establish
CTO	Consent to operate
DPR	Detailed Project Report
DR	Document Review
DISCOM	Distribution Company
EF	Emission Factor
EIA	Environmental Impact Assessment
ESIA	Environmental and Social Impact Assessment
EPC	Equipment procurement Contract
ER	Emission Reductions
FAR	Forward Action Request
FIFO	First In First Out
GHG	Greenhouse gas(es)
GS4GG	Gold Standard for Global Goals
GWP	Global Warming potential
LSC	Local Stakeholder Consultation
MPCB	Maharashtra Pollution Control Board
MoEF&CC	Ministry of Environment, Forest & Climate Change
MSW	Municipal Solid Waste
MW	Mega Watt
OM	Operating Margin
PCB	Pollution Control Board
PD	Project Proponent
PP	Project proponent
PPP	Public-Private Partnership
RBI	Reserve Bank Of India
RCC	Reinforced Cement Concrete
RDF	Refuse-Derived Fuel
SDG	Sustainable Development Goals
SEB	State Electricity Board
TPD	Tonnes per Day
ULB	Urban Local Body
VCS	Verified Carbon Standard
VCU	Verified Carbon Units
VVB	Validating and verifying Body

WTE	Waste to Energy
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APPENDIX 4: FINDINGS OVERVIEW

Table 1. CL from this validation

CL ID	01	Section no.	3.1	Date: 30-March-2024
Description of CL				
1. PP to clarify that leachate is not generated at the project site and if generated, how it is handled. 2. In the submitted PD version 01 dated 22-January-2024, in section 1.11, under the mentioned technical specifications, PP has mentioned the manufacturer as 'Hitachi Zosen India Pvt. Ltd. (HZIPL)' for Incinerator/Furnace. PP to clarify whether the manufacturer is same for all the components, if not, PP to provide the manufacturer of all the components. 3. PP to clarify more about leachate treatment plant (i.e. capacity, treatment process, O&M Process) mentioned in project report.				
Project proponent response				Date: 15-May-2024
1. The amount of leachate generated during the operation has been sent to PCMC leachate treatment plant. Hence, treatment of leachate does not come under scope of Antony Lara Renewable Energy Pvt Ltd. 2. The technical specification for the incinerator/Furnace has mentioned as 'Hitachi Zosen India Pvt. Ltd. (HZIPL)' in the 1.12 section of the updated PD apart from that there is another manufacturer ISGEC heavy engineering limited specified in the service agreement. 3. Leachate treatment plant has been clarified in the 1.12 section of the PD.				
Documentation provided by project proponent				
Leachate treatment plant process flow chart				
Manufacturer specification				
Revised PD				
VVB assessment				Date: 20-May-2024
1. PP has clarified that, leachate created during operations of project activity will be sent to PCMC's treatment facility along with the flow diagram of leachate treatment process. But there is no such evidence/contract/agreement that PCMC is responsible for leachate treatment. Hence this part of finding is open. 2. PP has submitted the revised PD to VVB team. During assessment VVB team observed that, PP has mentioned the manufacturer of each component but supporting document to verify the same found missing. Hence this part of finding is open. 3. PP has submitted the revised PD to VVB team. During assessment VVB team observed that, PP has clarified about responsibility of leachate treatment is of PCMC. Also, provided the process flow diagram of leachate treatment facility having capacity of 100 KLD in section 1.12 of revised PD. Hence this part of finding is closed.				
Project proponent response				Date: 15-May-2024
1. In comparison with the baseline scenario, the amount of leachate generated due to project activity is very minimal as the operation of plant involves majorly RDF which has more combustible characteristics rather than biodegradable nature. Hence, the emissions due to leachate generated has not been considered under project emissions. And the minimal amount of leachate produced from the project activity has been sent to Leachate Treatment Plant which is in the scope of PCMC and does not come under the scope of Project proponent. 2. The attested document with the details of each component has been submitted.				
Documentation provided by project proponent				Date:
Attested Technical specification document.				
VVB assessment				Date: 09-September-2024
1. VVB team confirmed the leachate treatment scenario during an onsite visit, noting that the leachate generated by the project activity is minimal. This is primarily because the plant's operation focuses on Refuse-Derived Fuel (RDF), which is largely combustible and less biodegradable. As a result, emissions from the leachate have not been considered within the project's emissions. Furthermore, the minimal				

leachate produced is sent to the Leachate Treatment Plant, which falls under the scope of PCMC and is not within the Project proponent's responsibility. Hence this part of finding is closed.
2. The Project proponent (PP) submitted the techno-commercial offer for a 14 MW plant dated 10-January-2018 as a reference for the technical specifications. These specifications, along with other relevant details, were reviewed during the onsite visit and subsequently confirmed by the VVB team. Hence this part of finding is closed.

CL ID	02	Section no.	3.2.1	Date: 30-March-2024
Description of CL				
PP to clarify the following thing observed during desk review;				
1. MPCB consent letter validity date till 31/10/2022 only. Latest document needs to be provided.				
2. MPCB mentioned capacity 100 MT/Day but in report 700 MT/Day.				
3. EIA report doesn't have date of preparation or submission.				
4. Building permission letter doesn't have owners name and site address (Columns are blank).				
5. Factory License and building permission letter is inconsistent with Plot number (Gut Number).				
Project proponent response				Date: 15-May-2024
1. Consent to operate letter dated 11/07/2023 from MPCB with a validity for a period up to 30/06/2026 has been submitted.				
2. In line with the concession agreement, the MPCB has acknowledged the waste treatment plant capacity as 1000 MT/day out of which 700 MT/day are being processed as RDF for WTE plant.				
3. As per the EIA notification 2006, Environmental Clearance is not required to be conducted for Power plant up to 15 MW, based on non-hazardous municipal				
1) waste power				
2) (Source: http://www.environmentwb.gov.in/pdf/EIA%20Notification,%202006.pdf). Hence, EIA study has not been conducted.				
4. The Building permission approval has been authorized by PCMC for 'Whole Moshi Dump area'. Hence, Building permission letter doesn't have owners name and site address.				
5. The Building permission approval and MPCB authorization letter have been authorized by PCMC for "Whole Moshi Dump area". Hence, both the authorized document consist the information of other waste treatment projects located in Moshi dump. Whereas the Factory License is applicable for the Antony Lara Waste to Energy Plant, therefore the building permission approval consist of additional gut number.				
Documentation provided by project proponent				
Consent to Operate letter				
Building permission approval				
VVB assessment				Date: 20-May-2024
1. PP has submitted the latest consent to operate letter date 11- July- 2023 from Maharashtra Pollution control board with application No. MPCB-CONSENT-0000165106 having validity till 30 - June - 2026 which covered the start date and some period of crediting period. Hence this part of finding is closed.				
2. PP has provided the latest consent report and revised PD to VVB team. During assessment VVB team observed that, Total project have capacity of 1000 MT from which 700 MT is waste to energy plant and PP got consent from pollution control board for the which is inline with project scenario. Hence this part of finding is closed.				
3. PP has submitted the weblink for the list of industries requiring EIA and those exempt from it to the VVB, as per the EIA Notification 2006, which was applicable at the time of project commissioning. The current project activity does not require an EIA, consistent with Schedule 1 (d) of the provided notification. Hence this part of finding is closed.				
4. PP has clarified that the building permission letter does not mention the owner's name because the land/relevant property belongs to the government and has been allocated for this project, which is being developed as per the requirements of the local government body. Hence this part of finding is closed.				

5. PP clarified the issue regarding the different gut numbers. The project is commissioned by PCMC on government land, while Antony Lara is a private entity. The gut number on the Factory License corresponds to the address of Antony Lara's industry, not the project activity. **Hence this part of finding is closed.**

CL ID	03	Section no.	3.2.1	Date: 30-March-2024
Description of CL				
PP to provide supporting documents for the Local stakeholder Consultation conducted for the project activity.				
Project proponent response				Date: 15-May-2024
Supporting documents for the Local stakeholder Consultation conducted for the project has been submitted.				
Documentation provided by project proponent				
Photographs, participant list, feedback forms				
VVB assessment				Date: 20-May-2024
PP has not yet provided the supporting documents I .e. Photographs, participant list, feedback forms to verify information mentioned under the PD. Hence this part of finding remains open.				
Project proponent response				Date: 11-August-2024
The supporting documents for LSC meeting mentioned as per PD has been submitted.				
Documentation provided by project proponent				
Photographs, participant list, feedback forms				
VVB assessment				Date:
PP has submitted the supporting document to VVB team. During assessment VVB team observed that, PP sent an invitation letter to the Municipal Commissioner on 25-October-2023 and to MPCB officials on 31-October-2023. An attendance sheet dated 03-November-2023, along with photographs and feedback forms from the local stakeholder consultation, was provided and found to be consistent with the requirements of the project activity. Hence comment is closed.				

CL ID	04	Section no.	3.3.2	Date: 30-March-2024
Description of CL				
1. It is not clear how PP will ensure that handling and storage of the collected waste near bunker does not result in anaerobic conditions and do not lead to further absorption of moisture. 2. PP shall clarify in the VCS PD, how project activity adheres to para 39 and 40 of the applicable methodology AMS-III.E.				
Project proponent response				Date: 15-May-2024
1. Handling and storage of the collected waste near bunker does not undergo anaerobic condition as the continuous mixing of waste takes place to maintain homogeneity and to avoid moisture absorption. In addition, the accumulated waste prior to current operation day will be scheduled for incineration to avoid any anaerobic degradation in the accumulated waste. 3) 2. In line with paragraph 39 the Quantity of waste gassified (Q_y), The quantity of auxiliary fuel used (Q_{fuel}) and Quantity of RDF/SB combusted ($Q_{non-biomass}$) has been monitored and incorporated in the 5.2 section of the PD. As per the para 40, “the project proponent shall demonstrate annually, through the assessment of common practices at proximate waste disposal sites”. Based on the Clarification No SSC 638 on the monitoring of the baseline for biomass waste disposal under AMS-III.E, “the paragraph 40 is applicable to those project activities for which it was demonstrated during validation that the newly				

generated waste treated in the project activity would have been disposed of at two or more distinct destinations or applications (e.g. procured by third parties, disposed in sites with distinct MCF values, used in agricultural applications, etc.)”. The project activity involves the waste being treated at a single destination in the baseline and there is no other destinations for waste disposal in the baseline scenario, there is no requirement of demonstration of common practice assessment annually, at proximate waste disposal sites.	
Documentation provided by project proponent	
Updated PD	
VVB assessment	Date: 20-May-2024
1. PP clarified the measures taken to prevent the generation of anaerobic conditions near the bunker. Continuous mixing of waste is conducted to maintain homogeneity and prevent moisture generation. Additionally, any waste accumulated prior to the current day of operation is scheduled for incineration to avoid anaerobic degradation. This was confirmed during the onsite visit. Hence this part of finding is closed. 2. PP has submitted the revised PD to the VVB team. During the assessment, the VVB team observed that PP provided all relevant parameters as per paragraph 39 of AMS.III-E under section 5.2 of the PD. PP also clarified the applicability of paragraph 40. Since the project activity involves waste being treated at a single destination in the baseline with no other destinations for waste disposal, there is no requirement for demonstrating common practice assessment annually at proximate waste disposal sites. This is in accordance with Clarification No SSC 638 on monitoring the baseline for biomass waste disposal under AMS-III.E. Hence this part of finding is closed.	

CL ID	05	Section no.		Date: 30-March-2024
Description of CL				
PP to submit following documents to verification team to verify the information provided in the VCS PD. 1. Copy of DPR/ Pre-Feasibility report of Project 2. Copy of Technical specification of equipment 3. Copy of Supporting for SDG's claimed 4. Copy of Power Purchase Agreement 5. Copy of JMR and Invoices on sample basis 6. Copy of No double counting of ACC Letter 7. Copy of Evidences for all the monitoring & supporting parameters				
Project proponent response				Date: 15-May-2024
1. Techno-commercial document of Project has been submitted. 2. Technical specification of equipment has been submitted. 3. In support of SDG waste water flow meter log book, employee list, pay slip details, waste and JMR have been submitted. 4. The conversion charges agreement entitled with the tariff detail has been submitted. 5. The Energy generation excel and Invoices on sample basis has been submitted. 6. The self-declaration form stating information of no double counting of VCUs has been submitted. 7. Copy of Evidences for all the monitoring parameters including log book details, calibration data, JMR have been submitted.				
Documentation provided by project proponent				
Technical specifications of Boiler, Incinerator, Flue gas treatment unit, ID Fan, Steam turbine, Generator, air cooled condenser etc Technical specification Employment records, salary slips Concession agreement and conversion charges agreement Energy generation records and Invoices No double counting declaration Meter calibration data				
VVB assessment				Date: 20-May-2024

1. PP has not provided the techno commercial document to VVB team. **Hence this part of finding is open.**
2. PP has submitted the service agreement document for technical specifications. During assessment VVB observed that, service agreement made between Hitachi Zosen India Pvt. Ltd. (HZIPL), ISGEC Heavy Engineering Ltd and Antony Lara Renewable energy Private Ltd. Dated 06/07/2020. Technical specifications mentioned in PD is found consistent with provided service agreement. **Hence this part of finding is closed.**
3. PP has submitted the leachate treatment facility details and supporting documents to support SDG 06, energy generation record for SDG 7, employee PF data for SDG 8, Amount of waste collected and processed data for SDG 11, Data of amount of waste utilised for energy generation from waste for SDG 12, and Estimated Energy generation sheet and relevant emission reduction for SDG 13 and concession agreement were provided for SDG 17. **Hence this part of finding is closed.**
4. PP has submitted the conversion charges agreement dated 26/06/2020 to VVB team. During assessment VVB team observed that, according to article 5 of agreement Antony Lara renewable energy Pvt. Ltd. Is responsible for energy supply till delivery point and after that PCMC will take care of supply to grid by its own and Antony Lara will get paid for same energy by monthly billing which found acceptable. PP also shares the monthly bills to VVB to verify the energy generation. **Hence this part of finding is closed.**
5. PP has provided excel of energy generation and monthly bills to VVB team. During assessment VVB team observed that, energy generation in excel and in provided monthly bills found consistent. **Hence this part of finding is closed.**
6. PP has not submitted the supporting document for No double counting of ACC Letter. **Hence this part of finding is open.**
7. PP needs to clearly mention the parameter and its relevant supporting document provided in response. **Hence this part of finding is open.**

Project proponent response

Date: 11-August-2024

1. The techno commercial document along with DPR has been submitted to VVB team.
6. Self-declaration on No double counting of VCU has been submitted
7. The monitoring parameter and its relevant supporting document has been submitted as given in below table.

	Monitoring parameter	Supporting document
1.	Quantity of net electricity generation supplied by the project	Energy generation reports
2.	Electricity Consumed by the project activity	Energy generation reports and Electricity invoices
3.	Total amount of waste disposed in a SWDS	Weigh bridge & Load cell records
4.	Weight fraction of the waste type	Waste characterization report
5.	Amount of waste combusted, gasified or mechanically/thermally treated by the project activity in each year y	Load cell - Plant Records
6.	Quantity of combustion residues generated by the project activity	Ash disposal record
7.	Quantity of non-biomass carbon content of the RDF combusted.	Waste records
8.	Quantity of auxiliary fossil fuel used	Auxiliary fossil fuel Record

6)

Documentation provided by project proponent	
Energy generation reports Electricity invoices Waste records Waste characterization report Load cell waste records Ash disposal record Auxiliary fossil fuel Record	
VVB assessment	Date:
1. The Project proponent (PP) submitted the techno-commercial offer for a 14 MW plant dated 10-January-2018 as a reference for the technical specifications. These specifications, along with other relevant details, were reviewed during the onsite visit and subsequently confirmed by the VVB team. Hence this part of finding is closed. 6. PP submitted a self-declaration letter dated 28 May 2024, confirming no double counting of ACCs generated. This declaration has been found acceptable for the project activity. Hence this part of finding is Closed. 7. PP has submitted the supporting documents of monitoring parameter to VVB team. During review VVB team observed that, Energy generation reports on sample basis from October 2023 to March 2024, Electricity invoices from October to December 2023, Weigh bridge & Load cell record of month from October 2023 to March 2024. Hence this part of finding is Closed.	

Table 2. CAR from this validation

CAR ID	01	Section no.	3.1	Date: 30-March-2024
Description of CAR				
1. PP need to submit the PD in latest version of PD available on VCS webpage. 2. In section 1.1, PP has not mentioned the estimated energy generation. 3. PP has applied VCS standard 4.5. However, on the VCS portal, latest version of VCS standard is 4.7. PP to provide updated version in PD. Correction is sought.				
Project proponent response				Date: 15-May-2024
1. PD has been updated in the latest version of template 4.3 available on VCS page. 2. In section 1.1, the estimated energy supplied to grid has been updated. 3. The latest version of VCS standard 4.7 has been applied and updated in PD				
Documentation provided by project proponent				
The revised PD				
VVB assessment				Date: 21-May-2024
1. PP has submitted the revised PD to VVB team. During assessment VVB team observed that, PP has used the updated version of PD i.e. version 4.3 which is applicable and found consistent with VEERA guidelines. Hence this part of finding is closed. 2. PP has submitted the revised PD to VVB team. During assessment VVB team observed that, PP has mentioned the energy generation of 77,616 MWh/Year which is as per template requirement and found consistent. Hence this part of finding is closed. 3. PP has provided revised PD to VVB team. During assessment VVB team observed that, PP has used the latest VCS standard Version 4.7 and same is mentioned revised PD. VVB confirmed that version 4.7 is the latest version on VEERA web page and found consistent. Hence this part of finding is closed.				

CAR ID	02	Section no.	3.3.5	Date: 30-March-2024
Description of CAR				

PP to provide justification on Project activity must not be mandatory by any law or regulatory framework is not provided.	
Project proponent response	Date: 15-May-2024
Justification on Project not mandatory by any law or regulatory framework has been provided under the section of 3.5 of the PD	
Documentation provided by project proponent	
Update PD	
VVB assessment	Date: 21-May-2024
PP has provided the revised PD to VVB team. During assessment VVB team observed that, PP has detailed all the legal requirements of the host country, confirming that there are no regulatory obligations for the development of this project. This has been verified by the VVB through relevant rules, regulations, and sectoral expertise. Hence this comment is closed.	

CAR ID	03	Section no.	3.3.6	Date: 30-March-2024
Description of CAR				
In the submitted PD version 01 dated 15-September-2023, in section 4, PP to mention the equation numbers as per the applied methodology. Correction is sought.				
Project proponent response				Date: 15-May-2024
All the equations in section 4 has been updated as per the applied methodology accordingly.				
Documentation provided by project proponent				
The revised PD				
VVB assessment				Date: 21-May-2024
PP has provided the revised PD to VVB team. During assessment VVB team observed that, PP has provided the relevant equation numbers in accordance with the methodologies and tools used. This has been verified against the relevant guidelines and found to be consistent. Hence this comment is closed.				

APPENDIX 5: ASSESSMENT TEAM

Name	SHORT CV. BACKGROUND INFORMATION
Mr. Deepak Pundlik	Mr. Deepak Pundlik has an experience in climate change, waste management and environmental management fields. He is a post graduate in Environment Sciences from Pune university, India. As a GHG consultant, he handled projects from renewable energy and waste management sectors during his stint with different companies like MITCON, Thermax Limited etc. As a GHG auditor, he is involved in validating/verifying projects under various carbon registries from renewable energy, energy demand, waste management sectors. Mr. Deepak Pundlik is based in Pune, India. Currently he is associated with True Quality Certifications Pvt. Ltd. (Applus+ Certification's Outsourced Entity). He is participating as a Lead auditor and Technical expert (1.1 and 13.1) in the project team.
Dr. Atul Takarkhede	Dr. Atul Takarkhede is Ph.D. (Environmental Sciences) from Institute of Science, RTM Nagpur University, Nagpur, and he has already published different technical papers related to environmental sciences. He counts with more than 11 years of experience in field of Environmental Auditing, consulting and accreditation. He is an expert in ISO 9001-14001, CO2/GHG Reporting, Carbon Foot Print, Energy, Water and Waste Management reporting for organizations' environmental performance. His professional portfolio is mainly related with carrying out EIA, conducting QA/QC of EIA Reports; conducting environmental/water audits; NABET requirements appliance, functional area expert in Water Pollution & Solid & Hazardous Waste management among others. Furthermore, he counts with solid experience on CDMVCS-GS consultancy and auditing. Currently he is associated with True Quality Certifications Pvt. Ltd. (Applus+ Certification's Outsourced Entity). Dr. Atul Takarkhede is based in Nagpur, India. Dr. Atul Takarkhede participates as part of the Audit Team as the Financial expert and Technical Expert (13.1) for the assessment.
Mr. Nikunj Agarwal	Mr. Nikunj Agarwal has very extensive experience (18 years) in the field of carbon market. He had been working with the different Europeans DoE's (Verification Bodies) accredited by UNFCCC for the Carbon Credits Certification. He had also worked as consultant for Energy Efficiency projects. Apparently, he had worked as Monitoring and Verification Experts in Energy Efficiency Projects. He had worked more than 300 Carbon projects under CDM/VCS/GS) in South Asia and others Region such as: India, Philippines, Malaysia, Fiji, Indonesia, China, Israel, Pakistan, Chile, Peru, Columbia, South Africa, Singapore, Nepal & Thailand. He is the Certified Energy Manager by the Bureau of Energy Efficiency, Government of India. He is in the Pool of Methodology Expert by the UNFCCC. He was part of the Gold Standard Advisory Panel for the CLIMATE SMART AGRICULTURE (CSA). Mr. Nikunj Agarwal is based in Kronberg, Germany. Mr. Nikunj Agarwal participates in the project's technical review team.